

A Project Report on

Solar-Based Wireless Charging For Electric Vehicles

Submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF TECHNOLOGY

In

ELECTRONICS AND COMMUNICATION ENGINEERING

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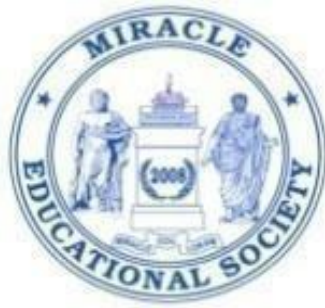
Department of Electronics and Communication Engineering
MIRACLE EDUCATIONAL SOCIETY GROUP OF INSTITUTION
(AUTONOMOUS)

(Affiliated to Jawaharlal Nehru Technology University Gurajada vizianagaram, **JNTUGV**)
Miracle city, Bhogapuram, Andhra Pradesh.

(2022-2026)

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CERTIFICATE

This is to certify that the project report entitled “**Solar-Based Wireless Charging Station For Electric Vehicles**” is submitted **Ch. Vandana (226C1A0410), K. Amrutha (226C1A0426), S. Navaneeth (226C1A0451), Ch. Abhishek (226C1A0411)**, in partial fulfilment of the requirements for the award of Degree of **Bachelor of Technology in Electronics and Communication Engineering from** MIRACLE Educational society Group of institutions Bhogapuram, affiliated to the Jawaharlal Nehru Technological University Gurajada Vizianagaram (**JNTUGV**), during the academic year 2020-2024, is a Bona fide work carried out by student under my guidance and supervision.

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DECLARATION

I hereby declare that the project report entitled “**Solar-Based Wireless Charging Station For Electric Vehicles**” submitted to the Jawaharlal Nehru Technological University Gurajada Vizianagaram (JNTUGV), is a record of an original work done by me under the guidance of **Dr. A.Venkateswar Rao**, Ph. D. Associate Professor in ECE Dept., and this project work is submitted in the partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in **Electronics and Communication Engineering**. The results shown in this report have not been submitted to any other University for the award of any degree.

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ACKNOWLEDGEMENT

I am extremely thankful to my Guide **Dr. A. Venkateswara Rao**, Associate Professor, Department of ECE, MIRACLE Educational society Group of institutions, Bhogapuram for their constant guidance, encouragement and moral support throughout the project. I am extremely grateful to **Mrs. D. Rajitha**, HOD, Department of ECE, MIRACLE Educational society Group of institutions, Bhogapuram for their inspiration and valuable guidance during entire course duration. I am extremely grateful to **Dr. A. ARJUNA RAO**, Principal, MIRACLE Educational society Group of institutions Bhogapuram for their inspiration and valuable guidance during entire course duration. I am extremely thankful to **Dr. A. Venkateswar Rao**, Associate Professor, Project coordinator, Department of ECE, for their constant guidance, encouragement and moral support throughout the project. I will be failing in duty if i do not acknowledge with grateful thanks to the management members by providing resources to carry out this project in MIRACLE Educational society Group of institutions, Bhogapuram. I express my thanks to all staff members and friends for all the help and co-ordination extended in bringing out this project successfully in time. Finally, I am very much thankful to my parents who guided me for every step towards success.

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ABSTRACT

The primary reason why people don't prefer electric vehicles is because of the unavailability of charging stations. Charging stations, unlike petrol bunks, aren't available everywhere. There always exists a fear as to what might happen if the vehicle runs out of battery. People are worried about more straightforward and faster commuting methods in our country rather than saving the Earth from the ill effects caused by pollution. The project mainly deals with a simple solution to make charging stations more accessible. The solution involves using public electricity and solar panels for the easy and hassle-free charging of Electric Vehicles.

The project is a device to transfer power wirelessly instead of using conventional copper cables and current carrying wires. The concept of wireless power transfer was introduced by Nikolas Tesla. This power is made to be transferred within a small range only for example charging rechargeable batteries etc. For demonstration purposes we have used LED'S that operates by using wireless power. This requires an electronic circuit for conversion of 12V Power supply. This project uses regulated 5V and 9V for the modules. 7805 three terminal voltage regulator and 7809 three terminal voltage regulators are used for voltage regulation. Therefore, by this way the power gets transferred through primary coil to secondary coil that are separated by certain distance around 3cm. Here the primary coil acts as transmitter and secondary coil receives the power to run a load. This project can be used to charge batteries from solar energy. Whenever vehicles enter into this station the sensors will be activated then automatically the vehicle gets charged. The information should be displayed in LCD.

Keywords: Electric Vehicles, Arduino nano, Solar panels.

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CHAPTER-1

INTRODUCTION TO EMBEDDED SYSTEM



1.1 INTRODUCTION:

An embedded system can be defined as a computing device that does a specific focused job. Appliances such as the air-conditioner, VCD player, DVD player, printer, fax machine, mobile phone etc. are examples of embedded systems. Each of these appliances will have a processor and special hardware to meet the specific requirement of the application along with the embedded software that is executed by the processor for meeting that specific requirement. The embedded software is also called “firm ware”. The desktop/laptop computer is a general purpose computer. You can use it for a variety of applications such as playing games, word processing, accounting, software development and so on. In contrast, the software in the embedded systems is always fixed listed below.

Embedded systems do a very specific task, they cannot be programmed to do different things. . Embedded systems have very limited resources, particularly the memory. Generally, they do not have secondary storage devices such as the CDROM or the floppy disk. Embedded systems have to work against some deadlines. A specific job has to be completed within a specific time. In some embedded systems, called real-time systems, the deadlines are stringent. Missing a deadline may cause a catastrophe-loss of life or damage to property. Embedded systems are constrained for power. As many embedded systems operate through a battery, the power consumption has to be very low. Some embedded systems have to operate in extreme environmental conditions such as very high temperatures and humidity.

1.2 OVERVIEW OF EMBEDDED SYSTEM ARCHITECTURE

Every embedded system consists of custom-built hardware built around a Central Processing Unit (CPU). This hardware also contains memory chips onto which the software is loaded. The software residing on the memory chip is also called the ‘firmware’.

The operating system runs above the hardware, and the application software runs above the operating system. The same architecture is applicable to any computer including a desktop computer. However, there are significant differences. It is not compulsory to have an operating system in every embedded system. For small appliances such as remote control units, air conditioners, toys etc., there is no need for an operating system and you can write only the software specific to that application. For applications involving complex processing, it is advisable to have an operating system. In such a case, you need to integrate the application software with the operating system and then transfer the entire software on to the memory chip. Once the software is transferred to the memory chip, the software will continue to run for a long time you don't need to reload new software.

Now, let us see the details of the various building blocks of the hardware of an embedded system. As shown in Fig. the building blocks are;

- Central Processing Unit (CPU)
- Memory (Read-only Memory and Random Access Memory)
- Input Devices
- Output devices
- Communication interfaces
- Application-specific circuitry

1.2.1 Central Processing Unit (CPU):

The Central Processing Unit (processor, in short) can be any of the following: microcontroller, microprocessor or Digital Signal Processor (DSP). A micro-controller is a low- cost processor. Its main attraction is that on the chip itself, there will be many other components such as memory, serial communication interface, analog-to digital converter etc. So, for small applications, a micro-controller is the best choice as the number of external components required will be very less. On the other hand, microprocessors are more powerful, but you need to use

many external components with them. DSP is used mainly for applications in which signal processing is involved such as audio and video processing.

1.2.2 Memory:

The memory is categorized as Random Access Memory (RAM) and Read Only Memory (ROM). The contents of the RAM will be erased if power is switched off to the chip, whereas ROM retains the contents even if the power is switched off. So, the firmware is stored in the ROM. When power is switched on, the processor reads the ROM; the program is executed.

1.2.3 Input / Output devices:

Unlike the desktops, the input devices to an embedded system have very limited capability. There will be no keyboard or a mouse, and hence interacting with the embedded system is no easy task. Many embedded systems will have a small keypad-you press one key to give a specific command. A keypad may be used to input only the digits. Many embedded systems used in process control do not have any input device for user interaction; they take inputs from sensors or transducers and produce electrical signals that are in turn fed to other systems.

The output devices of the embedded systems also have very limited capability. Some embedded systems will have a few Light Emitting Diodes (LEDs) to indicate the health status of the system modules, or for visual indication of alarms. A small Liquid Crystal Display (LCD) may also be used to display some important parameters.

1.2.4 Communication interfaces:

The embedded systems may need to, interact with other embedded systems as they may have to transmit data to a desktop. To facilitate this, the embedded systems are provided with one or a few communication interfaces such as RS232, RS422, RS485, Universal Serial Bus (USB), IEEE 1394, Ethernet etc.

Application-specific circuitry:

Sensors, transducers, special processing and control circuitry may be required for an embedded system, depending on its application. This circuitry interacts with the processor to carry out the necessary work.

1.3 CLASSIFICATION OF EMBEDDED SYSTEM:

Embedded systems are classified into four categories based on their performance and functional requirements:

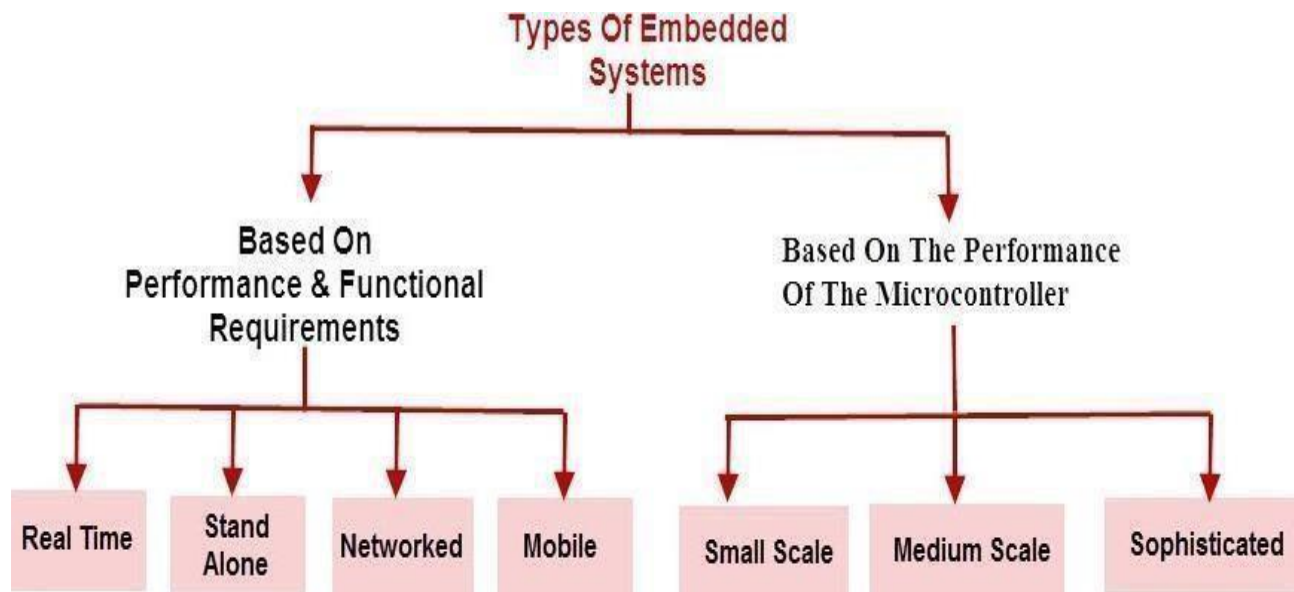


Fig 1.1: Classification of Embedded System

- Stand alone embedded systems
- Real time embedded systems
- Networked embedded systems
- Mobile embedded systems

Stand alone embedded system:

Stand alone embedded system don't require whole system like computer, it works by itself. It takes the input from the input ports either analog or digital and processes, calculates and converts the data and gives the result in data through the connected device-Which either controls, drives and displays the connected devices.



Fig 1.2: Standalone System

Real time embedded systems:

A real time embedded system is defined as, a system which gives a required output in a particular time. These type of embedded system follow the time dead lines for completion of a task.



Fig 1.3: Real time System

Networked embedded systems:

Network embedded system are related to a network access the resources. The connected network can be LAN ,WAN or the internet. The connected connection can be any wired or wireless.

Mobile embedded systems:

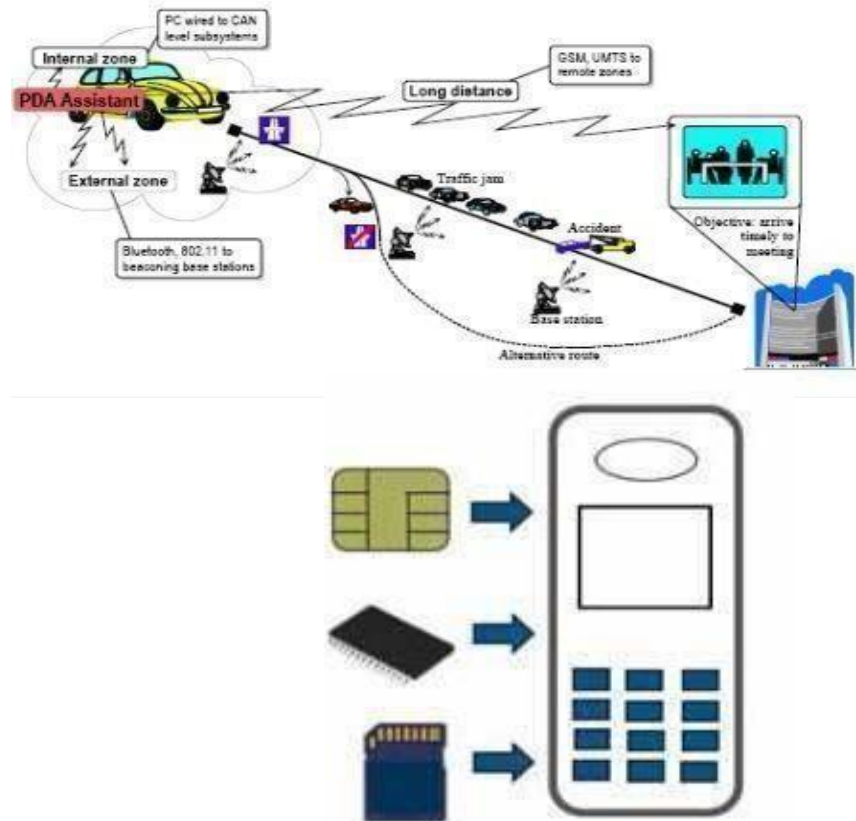


Fig 1.4: Mobile System

Mobile embedded systems are highly preferable in portable embedded devices like cell phones, mobiles, digital cameras, wireless mp3players and personal digital assistants etc. The basic limitations of this devices is the other resources and limitation of memory. Another category of embedded systems based on performance of micro controller and it is further classified into three categories as follows

Small scale embedded system:

This type of embedded systems are designed with a single 8-bit or 16-bit microcontroller. They have tiny scaled hardware, software complexities and involve board-level design. “C programming language” is used for developing these system.

Medium scale embedded systems:

These systems are usually designed with a single or few 16-bit or 32-bit microcontrollers or Digital Signal Processor or Reduced Instruction Set Computers being used. These systems have both hardware and software complexities.

Sophisticated embedded systems:

Sophisticated embedded systems have massive hardware and software complexities and may require ASIP's, IPs and PLAs scalable or configurable processors and programmable logic arrays.

1.4. APPLICATION AREAS

. Nearly 99 per cent of the processors manufactured end up in embedded systems. The embedded system market is one of the highest growth areas as these systems are used in very market segment- consumer electronics, office automation, industrial automation, biomedical engineering, wireless communication, data communication, telecommunications, transportation, military and so on.

Consumer appliances:

At home we use a number of embedded systems which include digital camera, digital diary, DVD player, electronic toys, microwave oven, remote controls for TV and air-conditioner, VCO player, video game consoles, video recorders etc. Today's high-tech car has about 20 embedded systems for transmission control, engine spark control, air-conditioning, navigation etc. Even wristwatches are now becoming embedded systems. The palmtops are powerful embedded systems using which we can carry out many general-purpose tasks such as playing games and word processing.

Office Automation:

The office automation products using embedded systems are copying machine, fax machine, key telephone, modem, printer, scanner etc.

Industrial automation:

Today a lot of industries use embedded systems for process control. These include pharmaceutical, cement, sugar, oil exploration, nuclear energy, electricity generation and transmission. The embedded systems for industrial use are designed to carry out specific tasks such as monitoring the temperature, pressure, humidity, voltage, current etc., and then take appropriate action based on the monitored levels to control other devices or to send information to a centralized monitoring station. In hazardous industrial environment, where human presence has to be avoided, robots are used, which are programmed to do specific jobs. The robots are now becoming very powerful and carry out many interesting and complicated tasks such as hardware assembly.

Medical Electronics:

Almost every medical equipment in the hospital is an embedded system. These equipments include diagnostic aids such as ECG, EEG, blood pressure measuring devices, X-ray scanners; equipment used in blood analysis, radiation, colonoscopy, endoscopy etc. Developments in medical electronics have paved way for more accurate diagnosis of diseases.

Computer Networking:

Computer networking products such as bridges, routers, Integrated Services Digital Networks (ISDN), Asynchronous Transfer Mode (ATM), X.25 and frame relay switches are embedded systems which implement the necessary data communication protocols. For example, a router interconnects two networks. The two networks may be running different protocol stacks. The router's function is to obtain the data packets from incoming ports, analyze the packets and send them towards the destination after doing necessary protocol conversion. Most networking equipments, other than the end systems (desktop computers) we use to access the networks, are embedded systems

Telecommunications:

In the field of telecommunications, the embedded systems can be categorized as subscriber terminals and network equipment. The subscriber terminals such as key telephones, ISDN phones, terminal adapters, web cameras are embedded systems. The network equipment includes

multiplexers, multiple access systems, Packet Assemblers Disassemblers (PADs), satellite modems etc. IP phone, IP gateway, IP gatekeeper etc. are the latest embedded systems that provide very low-cost voice communication over the Internet.

Wireless Technologies:

Advances in mobile communications are paving way for many interesting applications using embedded systems. The mobile phone is one of the marvels of the last decade of the 20th century. It is a very powerful embedded system that provides voice communication while we are on the move. The Personal Digital Assistants and the palmtops can now be used to access multimedia services over the Internet. Mobile communication infrastructure such as base station controllers, mobile switching centers are also powerful embedded systems.

Insemination:

Testing and measurement are the fundamental requirements in all scientific and engineering activities. The measuring equipment we use in laboratories to measure parameters such as weight, temperature, pressure, humidity, voltage, current etc. are all embedded systems. Test equipment such as oscilloscope, spectrum analyzer, logic analyzer, protocol analyzer, radio communication test set etc. are embedded systems built around powerful processors. Thank to miniaturization, the test and measuring equipment are now becoming portable facilitating easy testing and measurement in the field by field-personnel.

Security:

Security of persons and information has always been a major issue. We need to protect our homes and offices; and also the information we transmit and store. Developing embedded systems for security applications is one of the most lucrative businesses nowadays. Security devices at homes, offices, airports etc. for authentication and verification are embedded systems. Encryption devices are nearly 99 per cent of the processors that are manufactured end up in embedded systems.

Finance:

Financial dealing through cash and cheques are now slowly paving way for transactions using smart cards and ATM (Automatic Teller Machine, also expanded as Any Time Money) machines. Smart card, of the size of a credit card, has a small micro-controller and memory; and it interacts with the smart card reader! ATM machine and acts as an electronic wallet. Smart card technology has the capability of ushering in a cashless society. Well, the list goes on. It is no exaggeration to say that eyes wherever you go, you can see, or at least feel, the work of an embedded system.

Embedded systems find applications in . every industrial segment- consumer electronics, transportation, avionics, biomedical engineering, manufacturing, process control and industrial automation, data communication, telecommunication, defense, security etc. Used to encrypt the data/voice being transmitted on communication links such as telephone lines. Biometric systems using fingerprint and face recognition are now being extensively used for user authentication in banking applications as well as for access control in high security buildings.

CHAPTER-2

ANALYSIS AND DESIGN

2.1 Analysis of the work

The primary reason why people don't prefer electric vehicles is because of the unavailability of charging stations. Charging stations, unlike petrol bunks, aren't available everywhere. There always exists a fear as to what might happen if the vehicle runs out of battery. People are worried about more straightforward and faster commuting methods in our country rather than saving the Earth from the ill effects caused by pollution. The project mainly deals with a simple solution to make charging stations more accessible. The solution involves using public electricity and solar panels for the easy and hassle-free charging of Electric Vehicles.

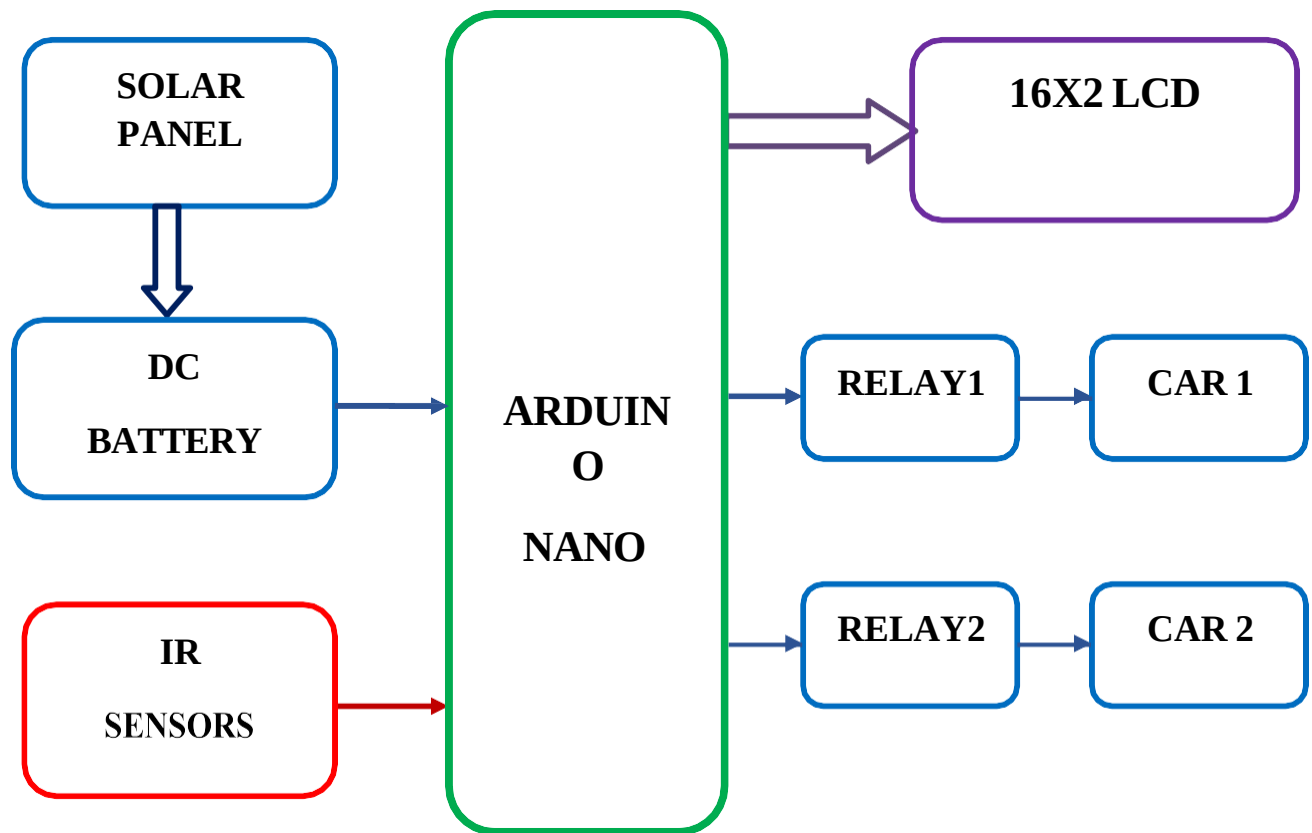


Fig 2.1: Block diagram of the System

Step-by-Step Charging Process

Sunlight to Energy Conversion: The solar panel converts sunlight into electricity. This energy is stored in the 12V battery.

Vehicle Detection Using IR Sensors:

When an electric vehicle enters the charging slot, the IR sensor detects it and sends a signal to the Arduino Nano.

Charging Activation via Relay:

Arduino activates the relay, allowing power to flow to the wireless charging coil.

Wireless Power Transfer:

The primary coil creates an electromagnetic field, and the secondary coil (inside the EV) picks up the energy.

Status Display on LCD:

The LCD screen updates the user with messages like:

“Vehicle Detected – Charging Started”

“Battery Full – Charging Stopped”

“Slot Available – Park to Charge”

Charging Completion:

Once charging is complete, the relay turns off to stop power transmission.

The LCD updates to indicate charging is done.

2.2 Hardware:

1. ARDUINO NANO (Atmega 328P)
2. SOLAR PANEL
3. 12V DC BATTREY
4. IR Sensors
5. 16x2 LCD
6. 2 channel Relay Module

2.3 Software:

1. Arduino IDE
2. Embedded c Language
3. Proteus Simulation Software

CHAPTER-3

Arduino Nano

3.1 Arduino Nano:

Arduino is an open-source hardware and software company, project and user community that designs and manufactures single-board microcontrollers and microcontroller kits for building digital devices and interactive objects that can sense and control objects in the physical and digital world.

The Arduino Nano is Arduino's classic breadboard friendly designed board with the smallest dimensions. The Arduino Nano comes with pin headers that allow for an easy attachment onto a breadboard and features a Mini-B USB connector.

The classic Nano is the oldest member of the Arduino Nano family boards. It is similar to the Arduino Duemilanove but made for the use of a breadboard and has no dedicated power jack. Successors of the classic Nano are for example the Nano 33 IoT featuring a WiFi module or the Nano 33 BLE Sense featuring Bluetooth® Low Energy and several environment sensors.

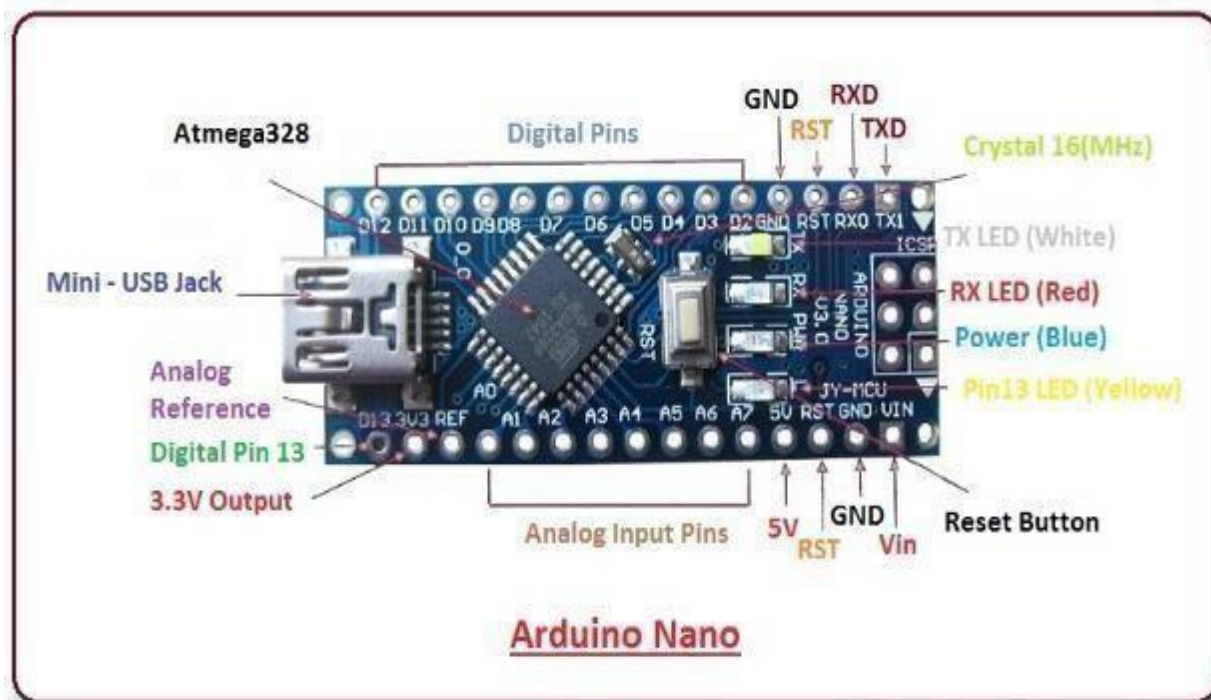


Fig 3.1: ARDUINO NANO

3.2 PIN DIAGRAM

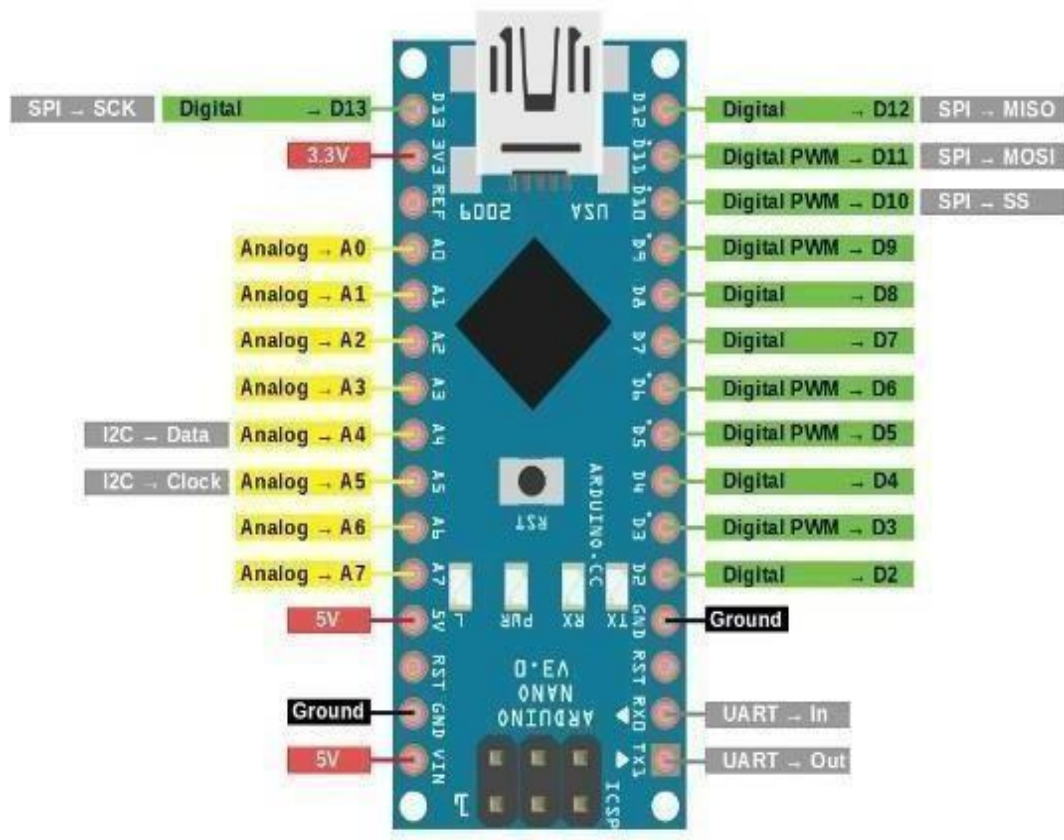


Fig 3.2 ARDUINO NANO PIN DIAGRAM:

The 14 digital input/output pins can be used as input or output pins by using `pinMode()`, `digitalRead()` and `digitalWrite()` functions in arduino programming. Each pin operate at 5V and can provide or receive a maximum of 40mA current, and has an internal pull-up resistor of 20-50 KOhms which are disconnected by default. Out of these 14 pins, some pins have specific functions as listed below:

- **Serial Pins 0 (Rx) and 1 (Tx):** Rx and Tx pins are used to receive and transmit TTL serial data. They are connected with the corresponding ATmega328P USB to TTL serial chip.
- **External Interrupt Pins 2 and 3:** These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value.
- **PWM Pins 3, 5, 6, 9 and 11:** These pins provide an 8-bit PWM output by using `analogWrite()` function.
- **SPI Pins 10 (SS), 11 (MOSI), 12 (MISO) and 13 (SCK):** These pins are used for SPI communication.
- **In-built LED Pin 13:** This pin is connected with an built-in LED, when pin 13 is HIGH – LED is on and when pin 13 is LOW, its off.

Along with 14 Digital pins, there are 6 analog input pins, each of which provide 10 bits of resolution, i.e. 1024 different values. They measure from 0 to 5 volts but this limit can be increased by using AREF pin with `analogReference()` function.

- Analog pin 4 (SDA) and pin 5 (SCA) also used for TWI communication using Wire library.
- Arduino Uno has a couple of other pins as explained below:
- **AREF:** Used to provide reference voltage for analog inputs with `analogReference()` function.
- **Reset Pin:** Making this pin LOW, resets the microcontroller.

3.3 SPECIFICATIONS:

Board	
Name	Arduino® Nano
SKU	A000005
Microcontroller	
ATmega328	

USB connector	
Mini-B USB	
Pins	
Built-in LED Pin	13
Digital I/O Pins	14
Analog input pins	8
PWM pins	6
Communication	
UART	RX/TX
I2C	A4 (SDA), A5 (SCL)
SPI	D11 (COPI), D12 (CIPO), D13 (SCK). Use any GPIO for Chip Select (CS).
Power	
I/O Voltage	5V
Input voltage (nominal)	7-12V
DC Current per I/O Pin	20 mA
Clock speed	
Processor	ATmega328 16 MHz
Memory	

ATmega328P	2KB SRAM, 32KB Flash, 1KB EEPROM
Dimensions	
Weight	7 g
Width	18 mm
Length	45 mm

CHAPTER-4

SOLAR PANEL AND BATTERY

4.1 SOLAR PANEL:

Solar PV modules are made by connecting photovoltaic (PV) cells or solar cells. They are made of semiconductor materials such as crystalline silicon. Solar modules convert light energy taken from the sun into electrical energy. So much energy generated by electricity is used to illuminate residential and commercial areas.

The Poly Crystalline Solar Panel 12V 20W is perfect for small battery charging. With its high efficiency, this solar panel is perfect for a variety of applications, including RVs, boats, cabins, and more. These Solar panels are a great way to invest in renewable energy. With a voltage of 12 V, these panels can be used in a variety of applications. The short circuit current is 1.45 Amp, making these panels ideal for a variety of needs. The voltage at max power is 18.20 V, and the current at max power is 1.1A.

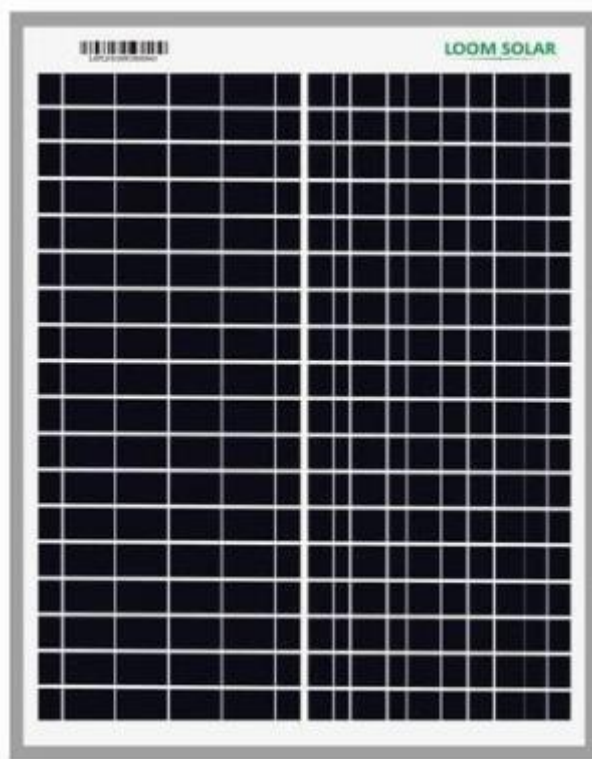


Fig:4.1 solar panel

Maximum power (pmax)	20 Wp
Maximum power voltage (Vmp)	19.25 V
Maximum power current (Imp)	1.04 A
Short circuit current (Isc)	1.11 A
Open circuit voltage (Voc)	22.5 V
Maximum system voltage	600 V

How do Solar PV Modules work?

A single PV cell is usually made of 12.5 cm and 15 cm square sizes and produces only 1 or 2 watts of energy. Solar PV modules are used to increase the power output of PV cells by connecting them. When PV cells (present in solar modules) absorb sunlight, the energy contained in light photons is transferred to semiconductor materials. Electrons are made to flow through semiconductor materials with an electric current. Typically, each solar PV module is rated at 50 W to 350 W. These modules can be connected or connected to an electrical grid to complete the solar PV system.

Technical Info

Nominal Maximum Power (Pm)	20Watts/12V
Open Circuit Voltage (Voc)	22.0V
Short Circuit Current (Isc)	1.2A
Voltage at Maximum Power (Vmp)	18.0V
Current at Maximum Power (Imp)	1.12A
Maximum System Voltage	600V

General Info

Length x Width x Thickness (LxWxT) – mm	520x355x22
Mounting Holes Pitch (Y) – mm	320
Mounting Holes Pitch (X) – mm	325
Solar Cells per Module (Units) / Arrangement	36 / (9 x 4)
Solar Cell Type	MultiCrystalline Silicon
Front Cover (Material / Thickness)	Tempered & Low Iron Glass / 3.2mm
Frame Material	Anodized Aluminum Alloy
Junction Box	IP67/IP68 Enclosure with Bypass Diodes

4.2 12V DC BATTERY:

The Amptek 12 Volt 1.3 Amp Rechargeable Battery offers a reliable power solution for various electronic devices and applications. With a capacity of 1.3 ampere-hours, this battery provides ample power to keep your devices running smoothly for extended periods. Whether you need backup power for security systems, portable electronics, or outdoor equipment, this rechargeable battery ensures dependable performance.

Equipped with a 12-volt output, the Amptek battery is compatible with a wide range of devices and systems. Its rechargeable design allows you to conveniently replenish its power supply, eliminating the need for frequent battery replacements and reducing waste. Whether you're powering LED lighting systems, portable fans, or small appliances, this battery delivers consistent and efficient performance.

The Amptek 12V 1.3 Amp Rechargeable Battery features a durable construction and reliable performance, making it ideal for both indoor and outdoor use. Its compact size and lightweight design make it easy to transport and install in various settings.

Technical Details

Manufacturer	expert power
Product Dimensions	15.24 x 9.65 x 9.4 cm; 3.45 Kilograms
Batteries	1 Lithium Ion batteries required. (included)
Item model number	12V 12AH
Colour	multicolour
Assembly Required	No
Voltage	12 Volts
Item Weight	3 kg 450 g

I2C LCD Display Driver:

I2C LCD Display Adapter is specially designed to drive LCD Display using only 2 data pins. This module uses the I2C serial bus to communicate with microcontroller like arduino. Normally it requires minimum 6 data pins to use a lcd display. It creates a problem when multiple sensor and this module is using with arduino at the same time. Because Arduino has limitation of I/O pins. This I2C LCD Display driver will help to reduce the uses of pins of your arduino and makes the project more easier. You can use this module with other I2C supported device on the same line and at the same time.

SPECIFICATIONS:

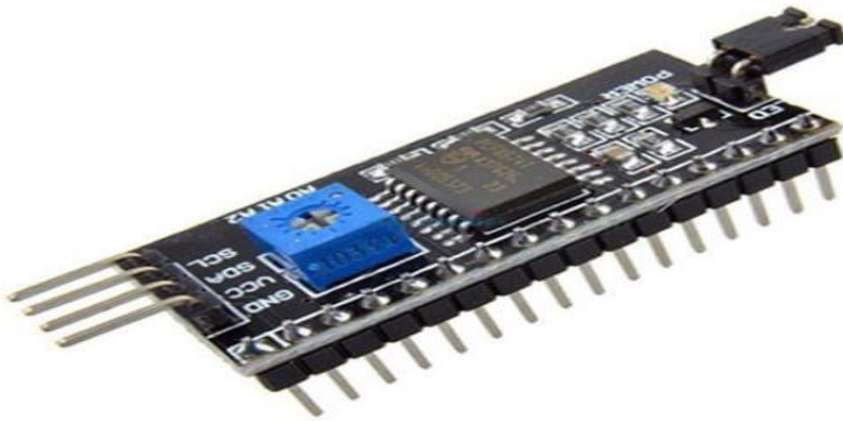
- I2C Address: 0x27 or 0x3F (Works any one of this 2 address)
- Operating Voltage: 5V
- Backlight can be turned on/off.
- Contrast can be changed using the onboard potentiometer.
- Total 16 Pins. Solder the 16 pins of driver with 16 pins of lcd.
- Compatible with both 16x2 and 20x4 Display.

CHAPTER-5

HARD WARE COMPONENTS

5.1 Alphanumeric LCD

Liquid Crystal Display also called as LCD is very helpful in providing user interface as well as for debugging purpose. The most commonly used Character based LCDs are based on Hitachi's HD44780 controller or other which are compatible with HD44580. The most commonly used LCDs found in the market today are 1 Line, 2 Line or 4 Line LCDs which have only 1 controller and support at most of 80 characters, whereas LCDs supporting more than 80 characters make use of 2 HD44780 controllers.



5.2 Pin Description

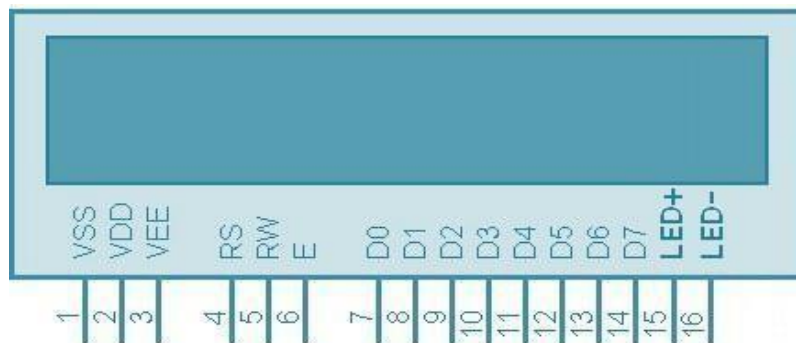


Fig: 5.1 LCD pin description

Pin No.	Name	Description
1	VSS	Power supply (GND)
2	VCC	Power supply (+5V)
3	VEE	Contrast adjust
4	RS	0 = Instruction input 1 = Data input
5	R/W	0 = Write to LCD module 1 = Read from LCD module
6	EN	Enable signal
7	D0	Data bus line 0 (LSB)
8	D1	Data bus line 1
9	D2	Data bus line 2
10	D3	Data bus line 3
11	D4	Data bus line 4
12	D5	Data bus line 5
13	D6	Data bus line 6
14	D7	Data bus line 7 (MSB)
15	LED+	Back Light VCC
16	LED-	Back Light GND

DDRAM - Display Data RAM

Display data RAM (DDRAM) stores display data represented in 8-bit character codes. Its extended capacity is 80 X 8 bits, or 80 characters. The area in display data RAM (DDRAM) that is not used for display can be used as general data RAM. So whatever you send on the DDRAM is actually displayed on the LCD. For LCDs like 1x16, only 16 characters are visible, so whatever you write after 16 chars is written in DDRAM but is not visible to the user.

CGROM - Character Generator ROM

Now you might be thinking that when you send an ASCII value to DDRAM, how the character is displayed on LCD? So the answer is CGROM. The character generator ROM generates 5 x 8 dot or 5 x 10 dot character patterns from 8-bit character codes. It can generate 208 5 x 8 dot character patterns and 32 5 x 10 dot character patterns.

CGRAM - Character Generator RAM

As clear from the name, CGRAM area is used to create custom characters in LCD. In the character generator RAM, the user can rewrite character patterns by program. For 5 x 8 dots, eight character patterns can be written, and for 5 x 10 dots, four character patterns can be written.

BF - Busy Flag

Busy Flag is a status indicator flag for LCD. When we send a command or data to the LCD for processing, this flag is set (i.e. $BF = 1$) and as soon as the instruction is executed successfully this flag is cleared ($BF = 0$). This is helpful in producing an exact amount of delay for the LCD processing.

To read Busy Flag, the condition $RS = 0$ and $R/W = 1$ must be met and The MSB of the LCD data bus (D7) act as busy flag. When $BF = 1$ means LCD is busy and will not accept next command or data and $BF = 0$ means LCD is ready for the next command or data to process.

5.3 Instruction Register (IR) and Data Register (DR)

There are two 8-bit registers in HD44780 controller Instruction and Data register. Instruction register corresponds to the register where you send commands to LCD e.g. LCD shift command, LCD clear, LCD address etc. and Data register is used for storing data which is to be displayed on LCD. When send the enable signal of the LCD is asserted, the data on the pins is latched in to the data register and data is then moved automatically to the DDRAM and hence is displayed on the LCD. Data Register is not only used for sending data to DDRAM but also for CGRAM, the address where you want to send the data, is decided by the instruction you send to LCD.

Commands and Instruction set

Only the instruction register (IR) and the data register (DR) of the LCD can be controlled by the MCU. Before starting the internal operation of the LCD, control information is temporarily stored into these registers to allow interfacing with various MCUs, which operate at different speeds, or various peripheral control

Command	Code										Description	Execution Time		
	RS	R/W	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0				
Clear Display	0	0	0	0	0	0	0	0	0	1	Clears the display and returns the cursor to the home position (address 0).	82μs~1.64ms		
Return Home	0	0	0	0	0	0	0	0	0	*	Returns the cursor to the home position (address 0). Also returns a shifted display to the home position. DD RAM contents remain unchanged.	40μs~1.64ms		
Entry Mode Set	0	0	0	0	0	0	0	0	1	I/D	S	Sets the cursor move direction and enables/disables the display.	40μs	
Display ON/OFF Control	0	0	0	0	0	0	0	1	D	C	B	Turns the display ON/OFF (D), or the cursor ON/OFF (C), and blink of the character at the cursor position (B).	40μs	
Cursor & Display Shift	0	0	0	0	0	1	S/C	R/L	*	*	Moves the cursor and shifts the display without changing the DD RAM contents.	40μs		
Function Set	0	0	0	0	1	DL	N\$	F	*	#	Sets the data width (DL), the number of lines in the display (L), and the character font (F).	40μs		
Set CG RAM Address	0	0	0	1	A _{CG}						Sets the CG RAM address. CG RAM data can be read or altered after making this setting.	40μs		
Set DD RAM Address	0	0	1	A _{DD}						Sets the DD RAM address. Data may be written or read after making this setting.	40μs			
Read Busy Flag & Address	0	1	BF	AC						Reads the BUSY flag (BF) indicating that an internal operation is being performed and reads the address counter contents.	1μs			
Write Data to CG or DD RAM	1	0	Write Data						Writes data into DD RAM or CG RAM.		46μs			
Read Data from CG or DD RAM	1	1	Read Data						Reads data from DD RAM or CG RAM.		46μs			
	I/D = 1: Increment S = 1: Accompanies display shift. S/C = 1: Display shift R/L = 1: Shift to the right. DL = 1: 8 bits N = 1: 2 lines F = 1: 5x10 dots BF = 1: Busy # Set to 1 on 24x4 modules \$ With KS0072 is Address Mode.										I/D = 0: Decrement S/C = 0: cursor move R/L = 0: Shift to the left. DL = 0: 4 bits N = 0: 1 line F = 0: 5 x 7 dots BF = 0: Can accept data		DD RAM: Display data RAM CG RAM: Character generator RAM A _{CG} : CG RAM Address A _{DD} : DD RAM Address Corresponds to cursor address. AC: Address counter Used for both DD and CG RAM address.	Execution times are typical. If transfers are timed by software and the busy flag is not used, add 10% to the above times.

devices. The internal operation of the LCD is determined by signals sent from the MCU. These signals, which include register selection signal (RS), read/write signal (R/W), and the data bus (DB0 to DB7), make up the LCD instructions (Table 3). There are four categories of instructions that:

- Designate LCD functions, such as display format, data length, etc.
- Set internal RAM addresses
- Perform data transfer with internal RAM
- Perform miscellaneous functions

Although looking at the table you can make your own commands and test them. Below is a brief list of useful commands which are used frequently while working on the LCD.

No.	Instruction	Hex	Decimal
1	Function Set: 8-bit, 1 Line, 5x7 Dots	0x30	48
2	Function Set: 8-bit, 2 Line, 5x7 Dots	0x38	56
3	Function Set: 4-bit, 1 Line, 5x7 Dots	0x20	32
4	Function Set: 4-bit, 2 Line, 5x7 Dots	0x28	40
5	Entry Mode	0x06	6
6	Display off Cursor off (clearing display without clearing DDRAM content)	0x08	8
7	Display on Cursor on	0x0E	14
8	Display on Cursor off	0x0C	12
9	Display on Cursor blinking	0x0F	15
10	Shift entire display left	0x18	24
12	Shift entire display right	0x1C	30
13	Move cursor left by one character	0x10	16

14	Move cursor right by one character	0x14	20
15	Clear Display (also clear DDRAM content)	0x01	1
16	Set DDRAM address or cursor position on display	0x80+add	128+add
17	Set CGRAM address or set pointer to CGRAM location	0x40+add	64+add

Sending Commands to LCD

To send commands we simply need to select the command register. Everything is same as we have done in the initialization routine. But we will summarize the common steps and put them in a single subroutine. Following are the steps:

- move data to LCD port
- select command register
- select write operation
- send enable signal
- wait for LCD to process the command

Sending Data to LCD

To send data we simply need to select the data register. Everything is same as the command routine. Following are the steps:

- move data to LCD port
- select data register

- select write operation
- send enable signal
- wait for LCD to process the data

LED: A light-emitting diode is a semiconductor light source that emits light when current flows through it. Electrons in the semiconductor recombine with electron holes, releasing energy in the form of photons. Typically, the forward voltage of an LED is between 1.8 and 3.3 volts. It varies by the color of the LED. A red LED typically drops around 1.7 to 2.0 volts, but since both voltage drop and light frequency increase with band gap, a blue LED may drop around 3 to 3.3 volts. The color of the light is determined by the energy required for electrons to cross the band gap of the semiconductor. White light is obtained by using multiple semiconductors or a layer of light-emitting phosphor on the semiconductor device. LEDs have many advantages over incandescent light sources, including lower power consumption, longer lifetime, improved physical robustness, smaller size, and faster switching. The disadvantages of LEDs include electrical limitations to low voltage and generally to DC (not AC) power.

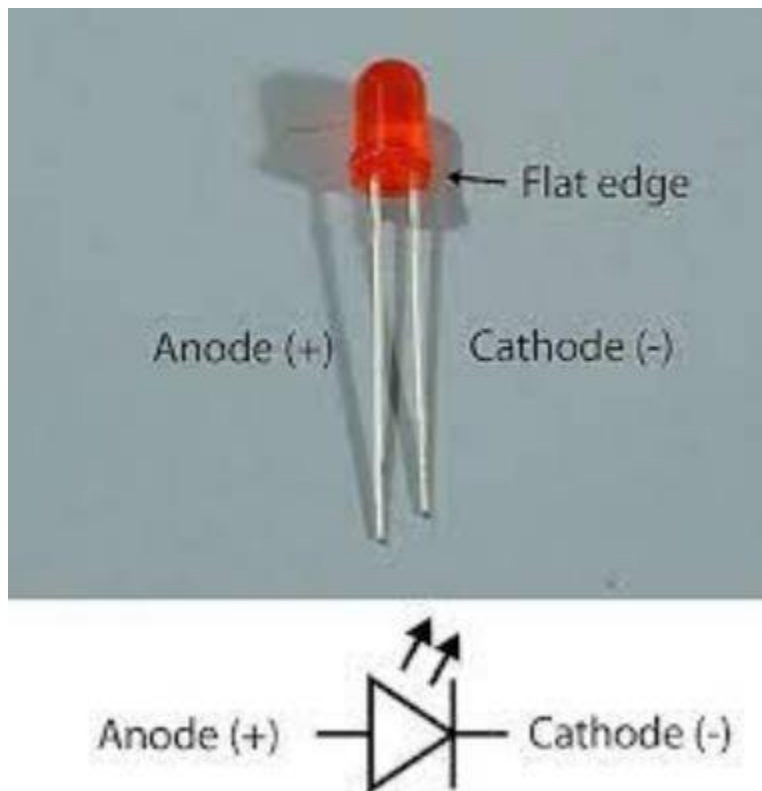


Fig: 5.2 LED image and Symbol

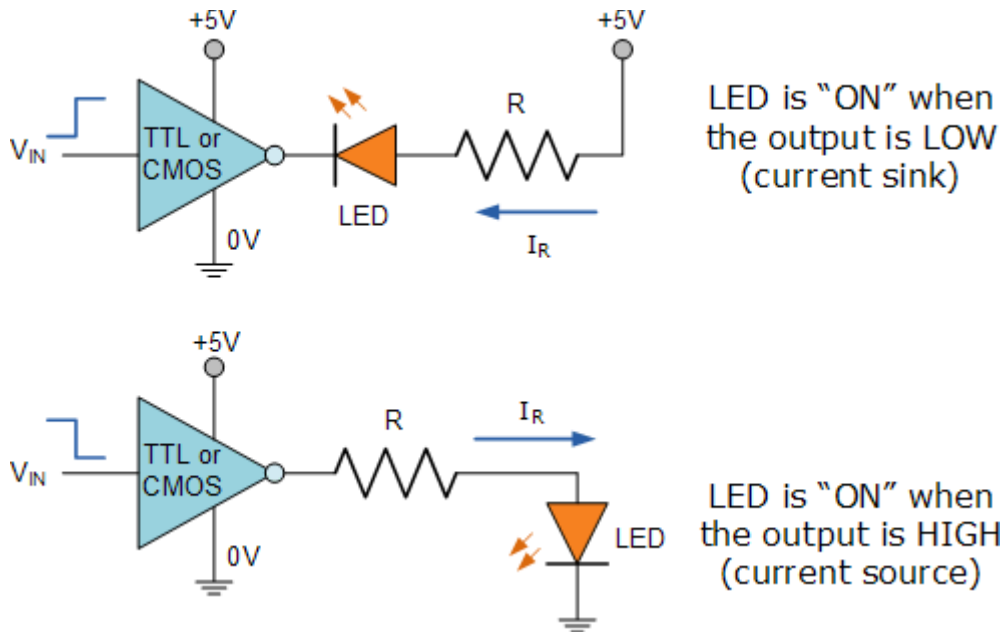


Fig: 5.3 LED Interfacing circuit

5.4 RELAY:

What is a relay?

We know that most of the high end industrial application devices have relays for their effective working. Relays are simple switches which are operated both electrically and mechanically. Relays consist of an electromagnet and also a set of contacts. The switching mechanism is carried out with the help of the electromagnet. There are also other operating principles for its working. But they differ according to their applications. Most of the devices have the application of relays.

Why is a relay used?

The main operation of a relay comes in places where only a low-power signal can be used to control a circuit. It is also used in places where only one signal can be used to control a lot of circuits. The application of relays started during the invention

of telephones. They played an important role in switching calls in telephone exchanges. They were also used in long distance telegraphy. They were used to switch the signal coming from one source to another destination.

Relay Design

There are only four main parts in a relay. They are

- Electromagnet
- Movable Armature
- Switch point contacts
- Spring

Relay Basics

The basics for all the relays are the same. Take a look at a 4 – pin relay shown below. There are two colours shown. The green colour represents the control circuit and the red colour represents the load circuit. A small control coil is connected onto the control circuit. A switch is connected to the load. This switch is controlled by the coil in the control circuit. Now let us take the different steps that occur in a relay.

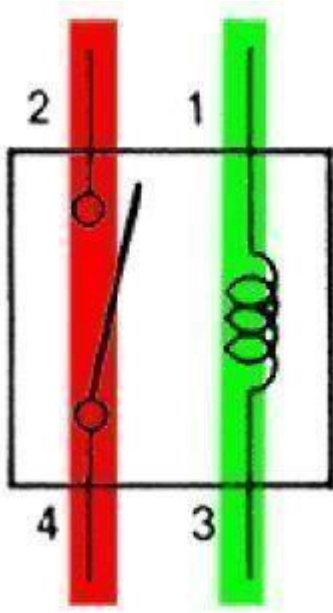


Fig 5.4 relay operation

- **Energized Relay (ON)**

As shown in the circuit, the current flowing through the coils represented by pins 1 and 3 causes a magnetic field to be aroused. This magnetic field causes the closing of the pins 2 and 4. Thus the switch plays an important role in the relay working. As it is a part of the load circuit, it is used to control an electrical circuit that is connected to it. Thus, when the relay in energized the current flow will be through the pins 2 and 4.

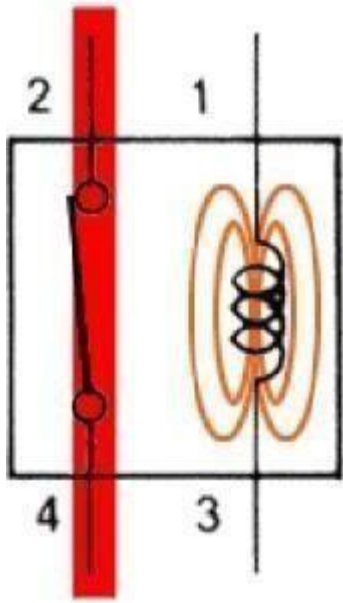


Fig 5.5 Energized Relay (ON)

- **De – Energized Relay (OFF)**

As soon as the current flow stops through pins 1 and 3, the switch opens and thus the open circuit prevents the current flow through pins 2 and 4. Thus the relay becomes de-energized and thus in off position.

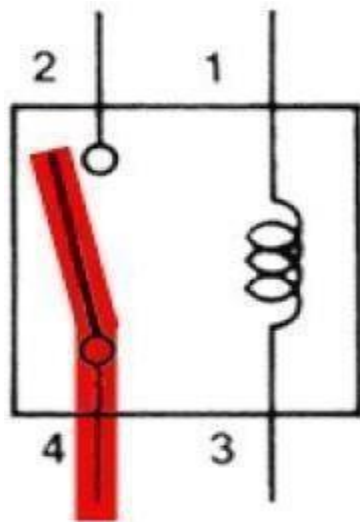


Fig 5.5.1 De-Energized Relay (OFF)

In simple, when a voltage is applied to pin 1, the electromagnet activates, causing a magnetic field to be developed, which goes on to close the pins 2 and 4 causing a closed circuit. When there is no voltage on pin 1, there will be no electromagnetic force and thus no magnetic field. Thus the switches remain open.

Pole and Throw

Relays have the exact working of a switch. So, the same concept is also applied. A relay is said to switch one or more poles. Each pole has contacts that can be thrown in mainly three ways. They are

- **Normally Open Contact (NO)** – NO contact is also called a make contact. It closes the circuit when the relay is activated. It disconnects the circuit when the relay is inactive.
- **Normally Closed Contact (NC)** – NC contact is also known as break contact. This is opposite to the NO contact. When the relay is activated, the circuit disconnects. When the relay is deactivated, the circuit connects.
- **Change-over (CO) / Double-throw (DT) Contacts** – This type of contacts are used to control two types of circuits. They are used to control a NO contact and also a NC contact with a common terminal. According to their type they are called by the names **break before make** and **make before break** contacts.

Relays are also named with designations like

- **Single Pole Single Throw (SPST)** – This type of relay has a total of four terminals. Out of these two terminals can be connected or disconnected. The other two terminals are needed for the coil.
- **Single Pole Double Throw (SPDT)** – This type of a relay has a total of five terminals. Out of these two are the coil terminals. A common terminal is also included which connects to either of two others.

- **Double Pole Single Throw (DPST)** – This relay has a total of six terminals. These terminals are further divided into two pairs. Thus they can act as two SPST's which are actuated by a single coil. Out of the six terminals two of them are coil terminals.
- **Double Pole Double Throw (DPDT)** – This is the biggest of all. It has mainly eight relay terminals. Out of these two rows are designed to be change over terminals. They are designed to act as two SPDT relays which are actuated by a single coil.

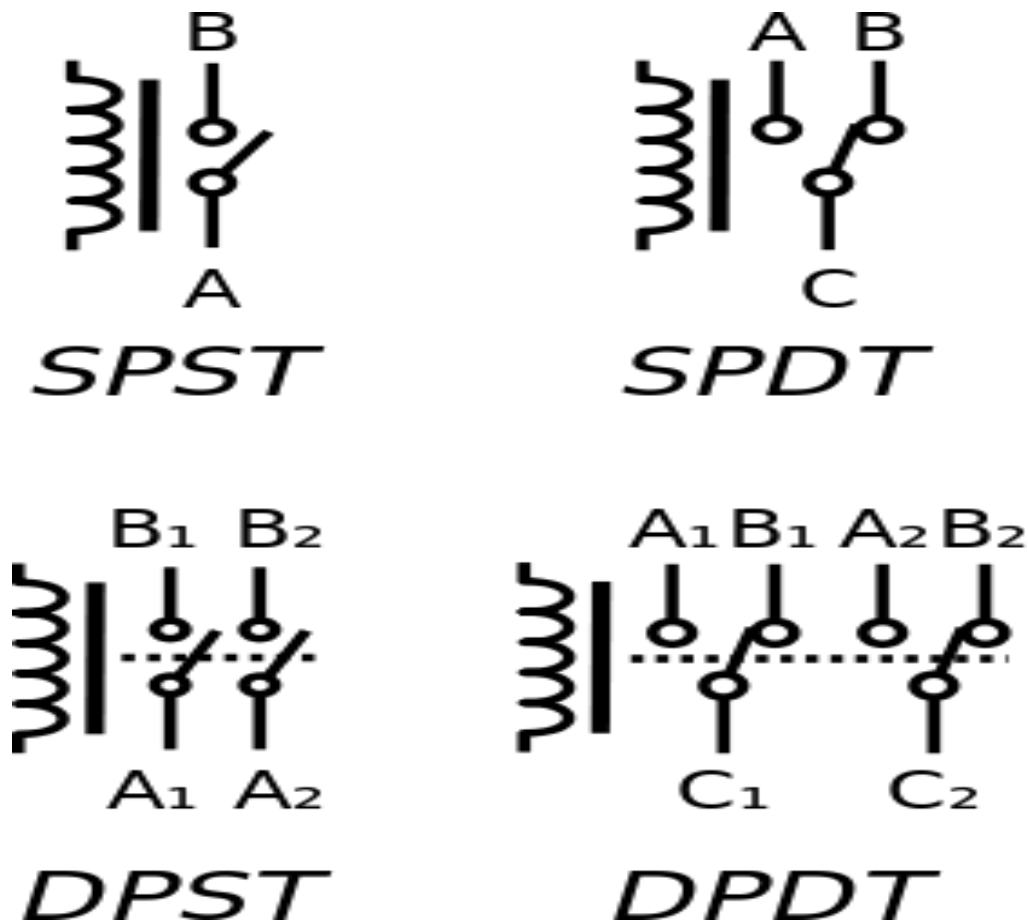


Fig 5.6 Two-Channel Relay Module Pinout:

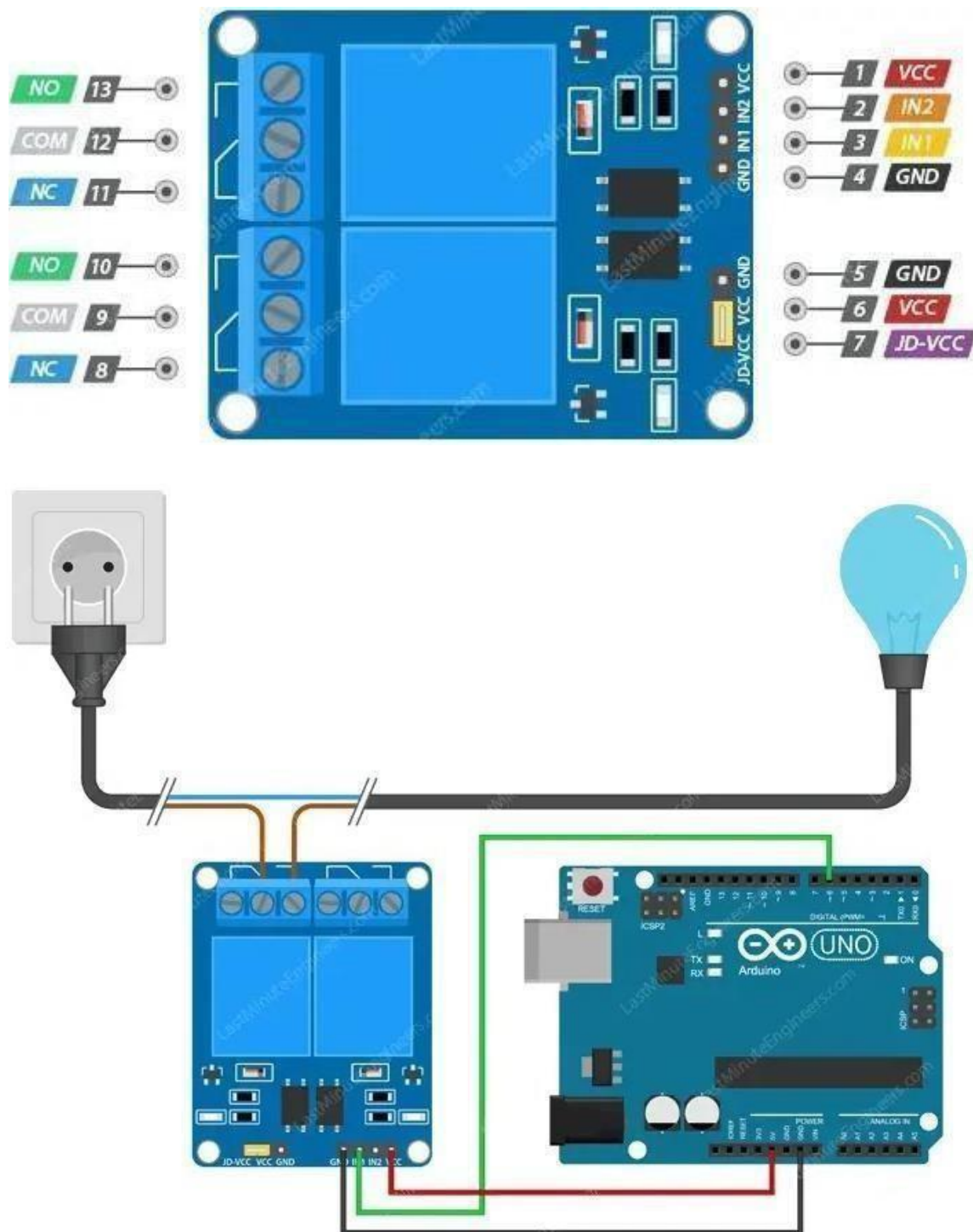


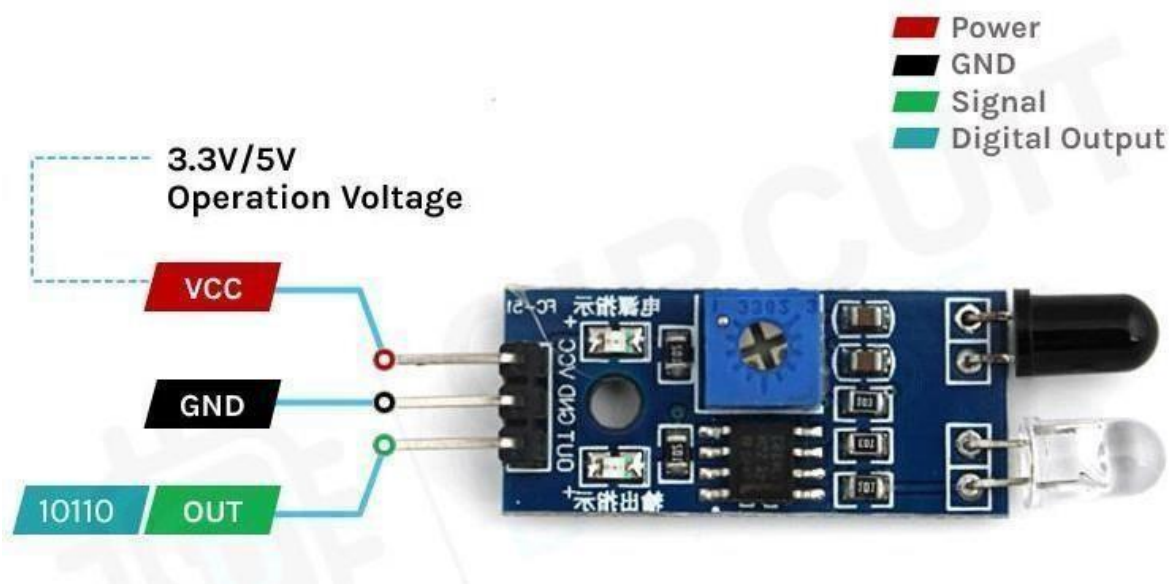
Fig 5.7 Arduino to sensor connection

5.5 IR SENSOR:

An **infrared proximity sensor or IR Sensor** is an electronic device that emits infrared lights to sense some aspect of the surroundings and can be employed to detect the motion of an object. As this is a passive sensor, it can only measure infrared radiation.

IR Sensor Pinout

The IR sensor has a 3-pin connector that interfaces it to the outside world. The connections are as follows:

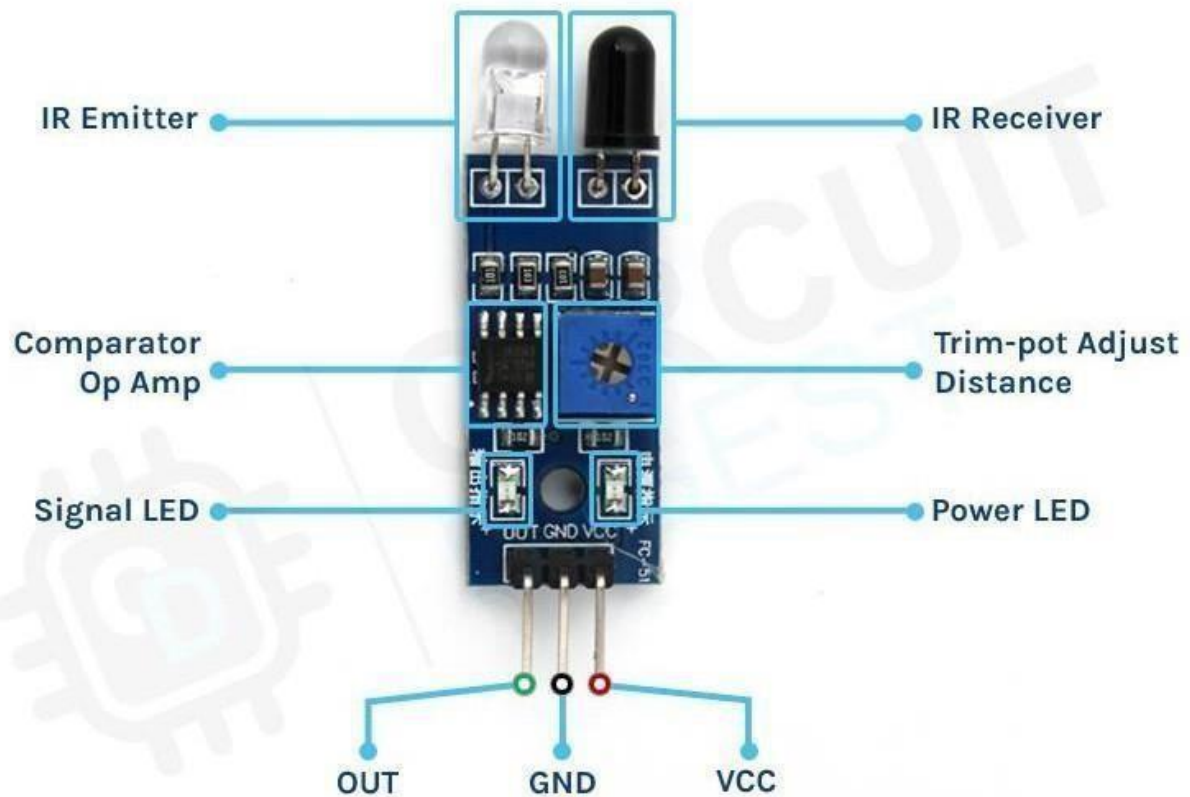


VCC is the power supply pin for the IR sensor which we connect to the 5V pin on the Arduino.

OUT pin is a 5V TTL logic output. LOW indicates no motion is detected; HIGH means motion is detected.

GND Should be connected to the ground of the Arduino.

This Sensor is popular among beginners as these are low power, low cost, rugged, and feature a wide sensing range that can be trimmed down to adjust the sensitivity.



How Does an IR Motion Sensor Module Work?

The **working of the IR sensor module** is very simple, it consists of two main components: the first is the IR transmitter section and the second is the IR receiver section. In the transmitter section, **IR led** is used and in the receiver section, a **photodiode** is used to receive infrared signal and after some signal processing and conditioning, you will get the output.

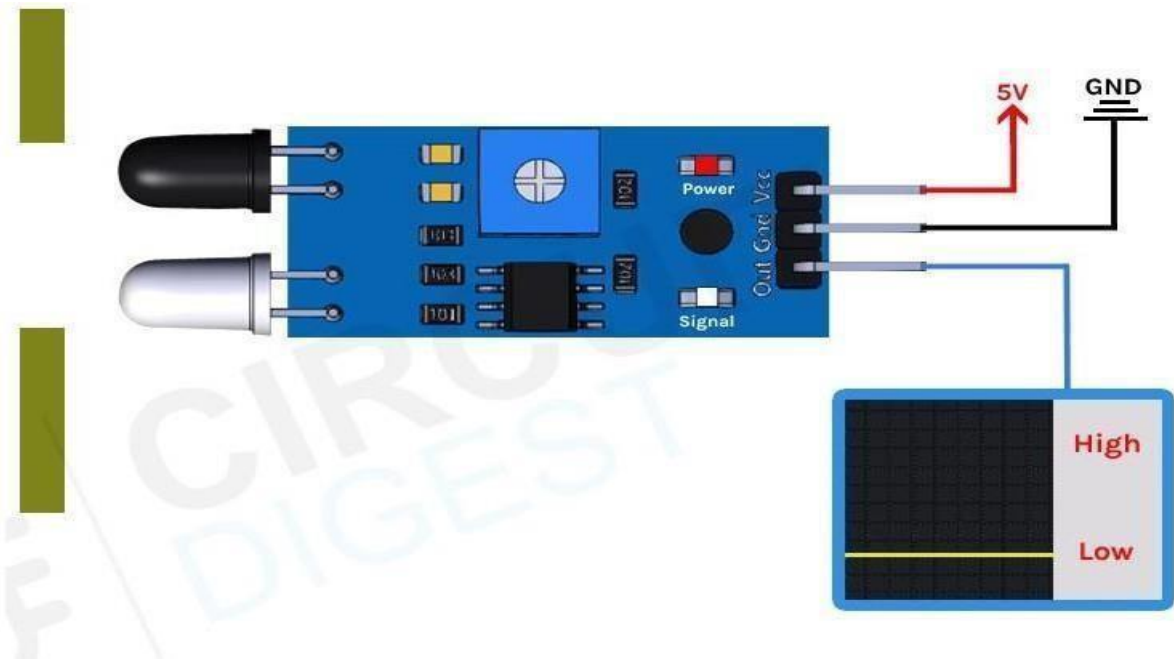


Fig:5.8 No obstacle Detected

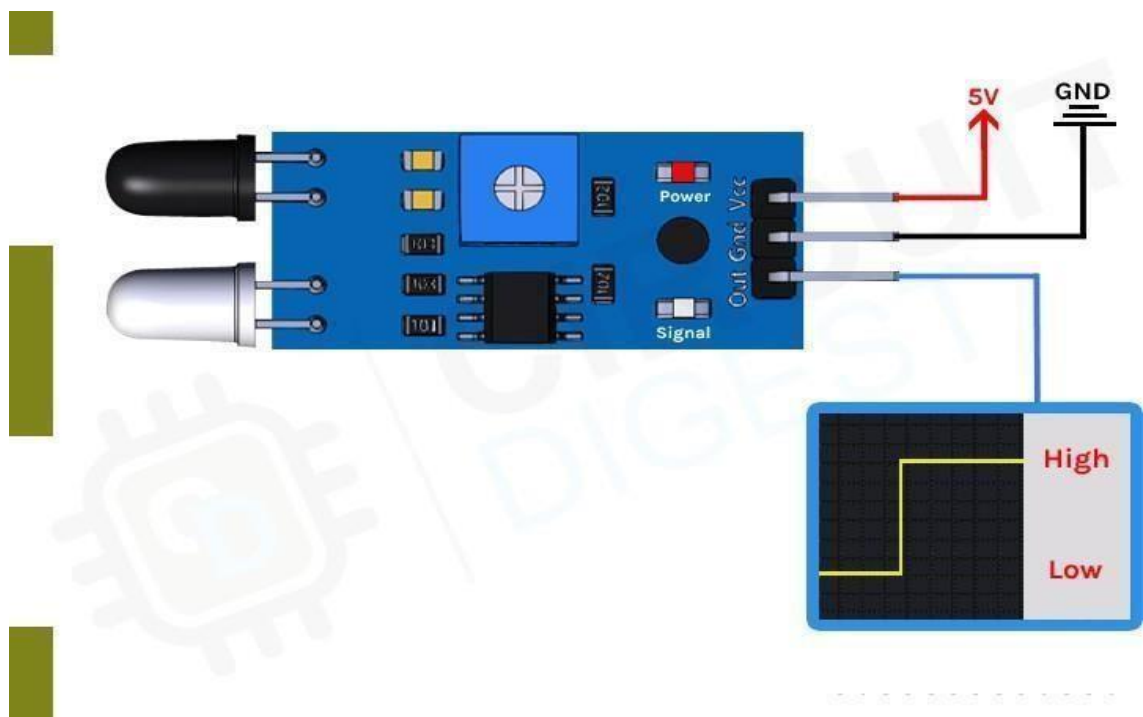


Fig: 5.9 obstacle Detected

An IR proximity sensor works by applying a voltage to the onboard **Infrared Light Emitting Diode** which in turn emits infrared light. This light propagates through the air and hits an object, after that the light gets reflected in the photodiode sensor. If the object is close, the reflected light will be stronger, if the object is far away, the reflected light will be weaker. If you look closely toward the module. When the sensor becomes active it sends a corresponding **Low signal** through the output pin that can be sensed by an Arduino or any kind of microcontroller to execute a particular task. The one cool thing about this module is that it has two onboard LEDs built-in, one of which lights on when power is available and another one turns on when the circuit gets triggered.

CHAPTER-6

SOFTWARE IMPLEMENTATION

SOFTWARE IMPLEMENTATION

6.1 ARDUINO IDE

To Program an Arduino:

Step1:

Arduino micro controllers come in a variety of types. The most common is the Arduino UNO, but there are specialized variations. Before you begin building, do a little research to figure out which version will be the most appropriate for your project.

Step2:

To begin, you need to install the Arduino Programmer, Integrated development environment (IDE).

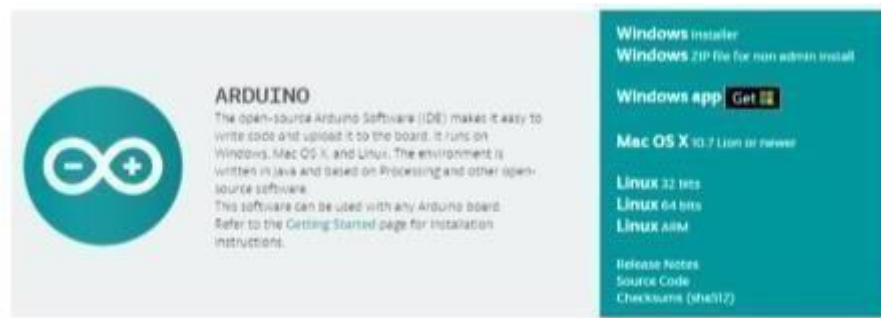


Fig: 6.1.1 How To Install Arduino IDE

To install the software, you will need to click on the link that corresponds with your computer's operating system. Once the software has been installed on your computer, go ahead and open it up. This is the Arduino IDE and is the place where all the programming will happen. Take some time to look around and get comfortable with it.

1. **Menu Bar:** Gives you access to the tools needed for creating and saving Arduino sketches.
2. **Verify Button:** Compiles your code and checks for errors in spelling or syntax.
3. **Upload Button:** Sends the code to the board that's connected such as Arduino Uno in this case. Lights on the board will blink rapidly when uploading.

4. **New Sketch:** Opens a new window containing a blank sketch.

5. **Sketch Name:** When the sketch is saved, the name of the sketch is displayed here.

6. **Open Existing Sketch:** Allows you to open a saved sketch or one from the stored examples.



Fig: 6.1.2 How To Start Arduino IDE

7. **Save Sketch:** This saves the sketch you currently have opened.

8. **Serial Monitor:** When the board is connected, this will display the serial information of your Arduino

9. **Code Area:** This area is where you compose the code of the sketch that tells the board what to do.

10. **Message Area:** This area tells you the status on saving, code compiling, errors and more.

11. **Text Console:** Shows the details of error messages, size of the program that was compiled and additional info.

12. **Board and Serial Port:** Tells you what board is being used and what serial port it's connected to.

Step 3:

Connect your Arduino to the USB port of your computer. This may require a specific USB cable. Every Arduino has a different virtual serial-port address, so you need to reconfigure the port if you are using different Arduino.



Fig 6.1.3: Connection of Arduino to PC

Step 4:

Once the board is connected, you will need to go to Tools then Board then finally selects Arduino nano.

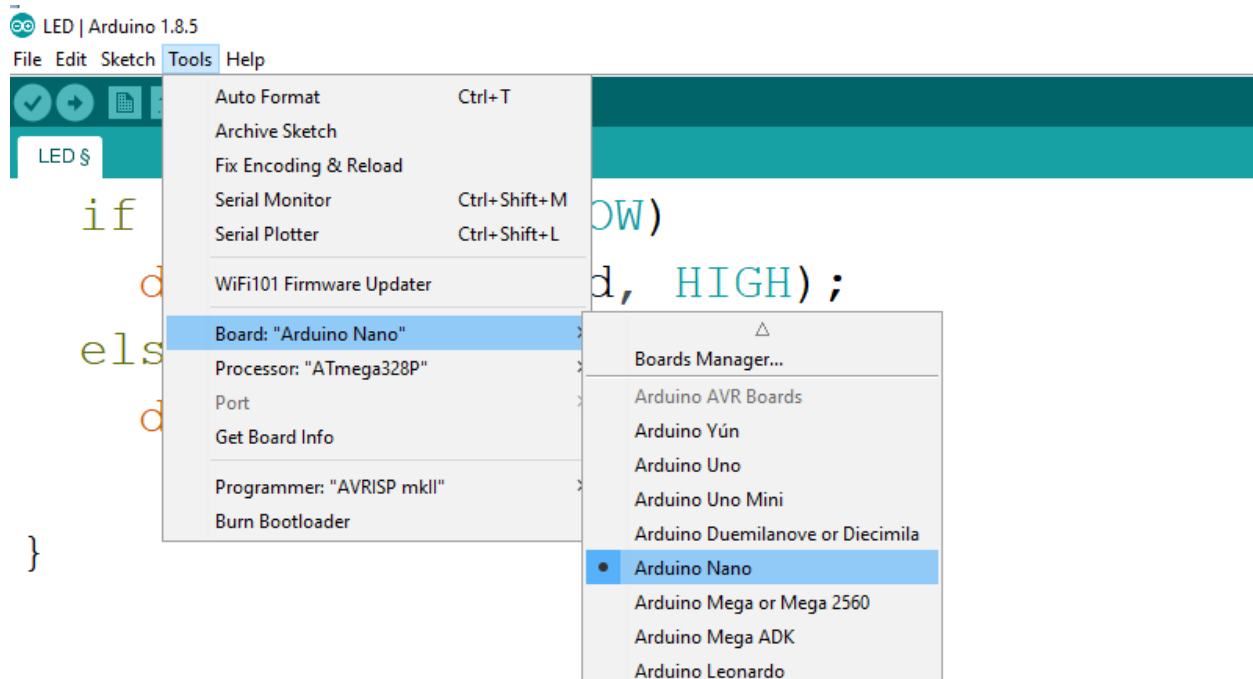


Fig: 6.1.4 Selection of Board

Next, you must tell the Arduino which port you are using on your computer. To select the port, go to Tools then Port then select the port that says Arduino.

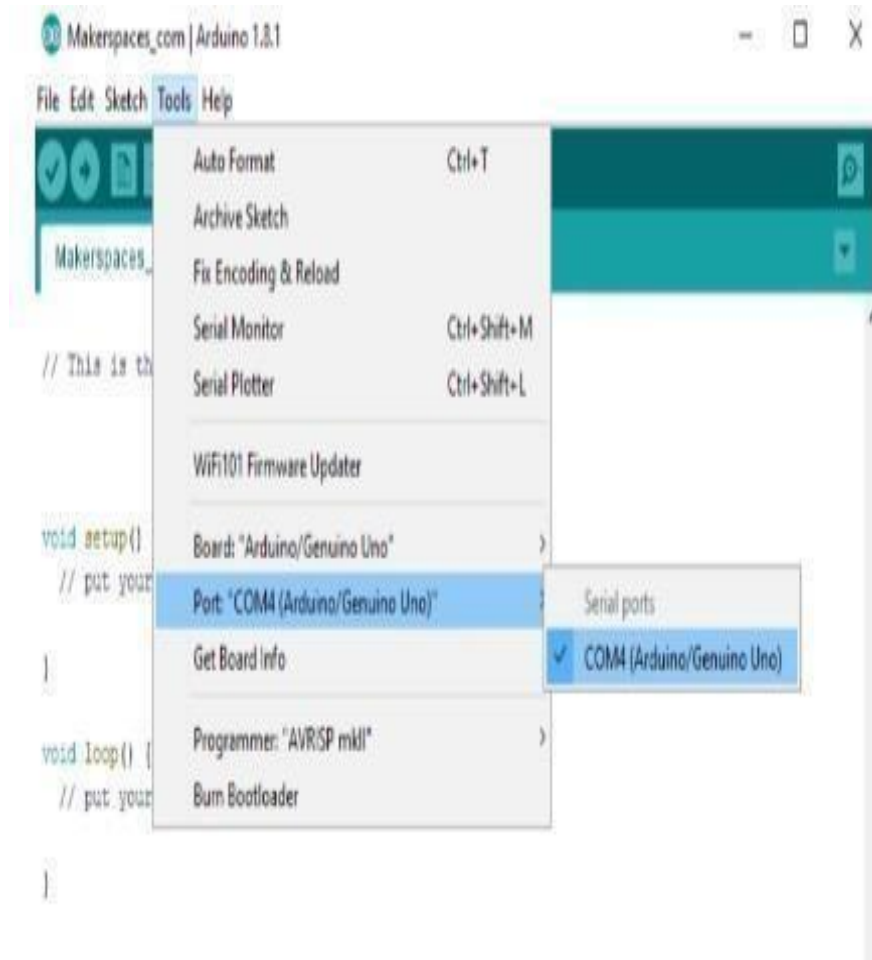


Fig: 6.1.5 Selection of Port

Step5:

Test the micro controller by using one of the preloaded programs, called sketches, in the Arduino Programmer. Open one of the examples sketches and press the upload button to load it. The Arduino should begin responding to the program: If you set it to blink an LED light, for example, the light should start blinking.

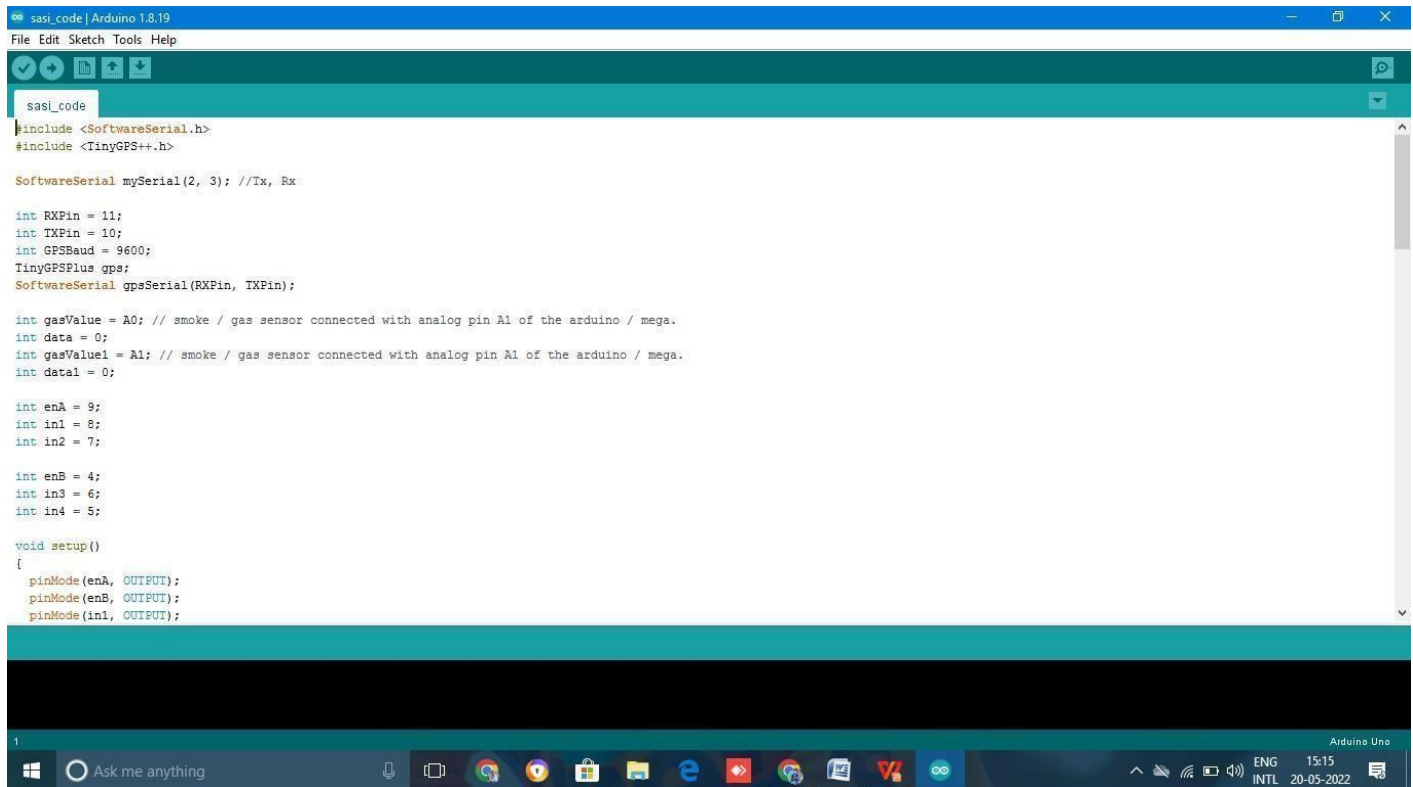


Fig: 6.1.6 Uploading A Program

The “void setup ()” section is widely used to initialize variables, pin modes, set the serial baud rate and related. The software only goes through the section once. The “void loop ()” section is the part of the code that loops back onto itself and is the main part of the code. In the Arduino examples, this is called “Bare Minimum” under File->Examples->Basics. Note that you are free to add subroutines using the same syntax:

Void subroutine name () {}

Almost every line of code needs to end with a semicolon ‘;’ (there are a few exceptions which we will see later). To write single line comments in the code, type two back slashes followed by the text:

//comments are overlooked when compiling your program

To write multi-line comments, start the comment with /* and end with */

/* this is a multi-line comment and saves you having to always use double slashes at the beginning of every line. Comments are used to explain the code textually. Good code always has a lot of comments. */

Step6:

To upload new code to the Arduino, either you need to have access to code you can paste into the programmer, or you must write it yourself, using the Arduino programming language to create your own sketch. An Arduino sketch usually has five parts: a header describing the sketch and its author; a section defining variables; a setup routine that sets the initial conditions of variables and runs preliminary code; a loop routine, which is where you add the main code that will execute repeatedly until you stop running the sketch; and a section where you can list other functions that activate during the setup and loop routines. All sketches must include the setup and loop routines.

Step 7

Once you've uploaded the new sketch to your Arduino, disconnect it from your computer and integrate it into your project as directed.

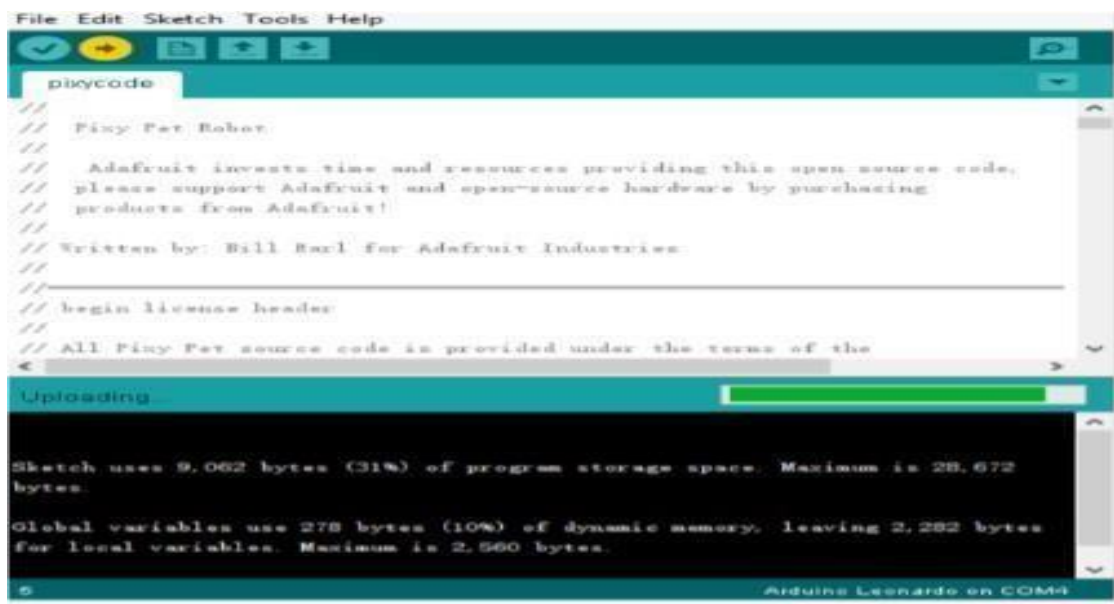


Fig: 6.1.7 To Upload A New Sketch

CHAPTER-7

RESULTS

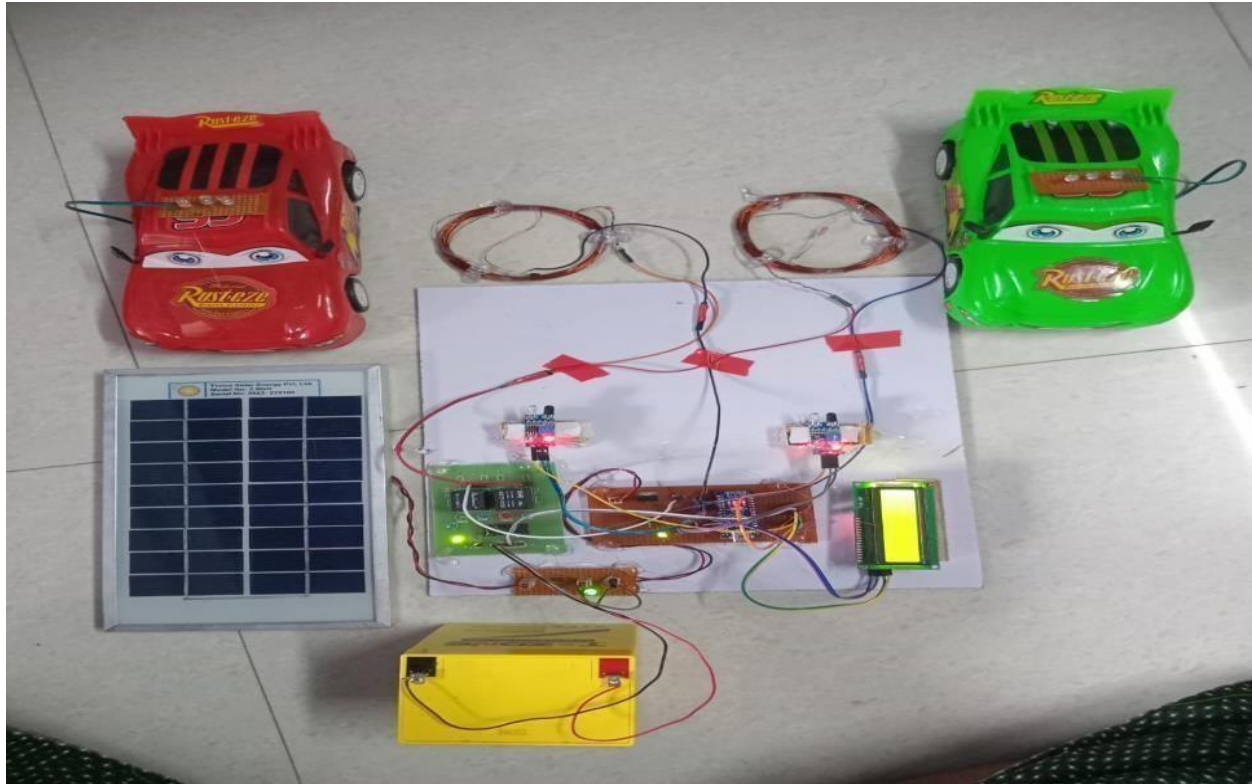


Fig 7.1: Complete setup for Realtime solar based charging system

- ***This image shows the hardware implementation of the solar-based wireless charging station.***
- ***Components visible in the setup include the solar panel, Arduino Nano, relay module, IR sensors, LCD display, and charging pads.***
- ***The system is designed to wirelessly charge electric vehicles using inductive power transfer, with the Arduino controlling the charging process.***

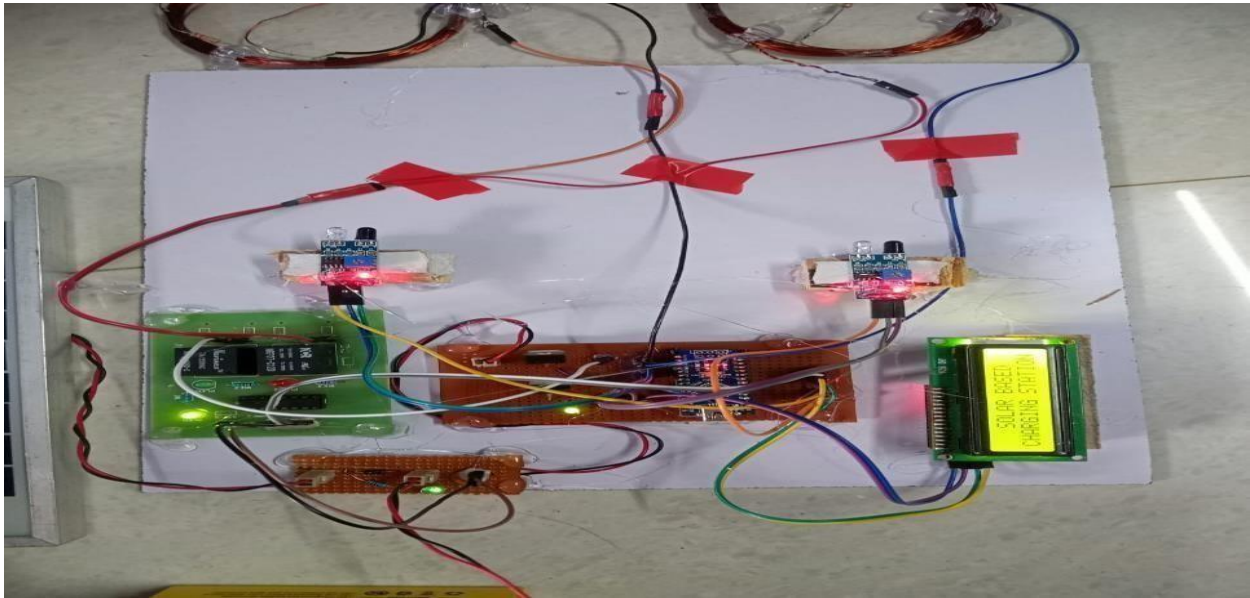


Fig 7.2: Real time view for solar based charging station

- *This diagram provides a real-world view of how the charging station operates.*
- *It shows the solar panel installed to capture sunlight and generate power for the system.*
- *Vehicles can be seen parked at dedicated charging slots, where wireless power transfer occurs when a car is detected.*
- *The LCD display updates the charging status, and the system operates automatically without human intervention.*

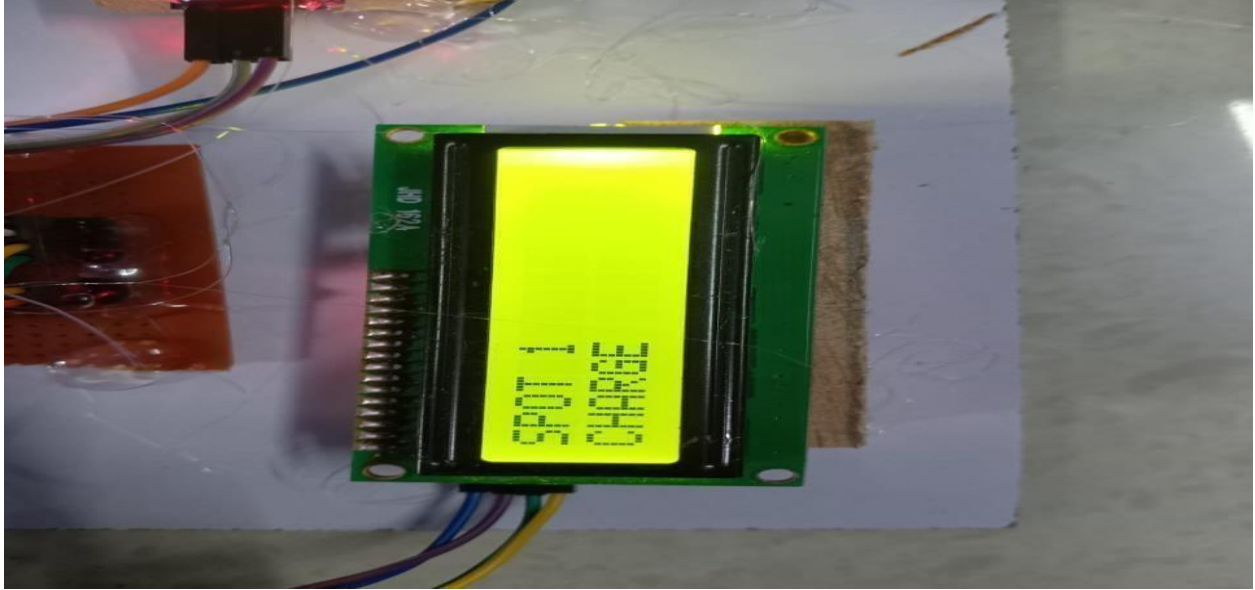


Fig 7.3: Detected single slot for charging the car

- *This image represents a scenario where one charging slot is occupied by an electric vehicle.*
- *The IR sensor detects the presence of a vehicle in Slot 1 and sends a signal to the Arduino Nano.*
- *The LCD display updates, showing that Slot 1 is occupied and charging is in progress.*
- *The relay switches ON, allowing power to flow to the charging pad for wireless energy transfer*

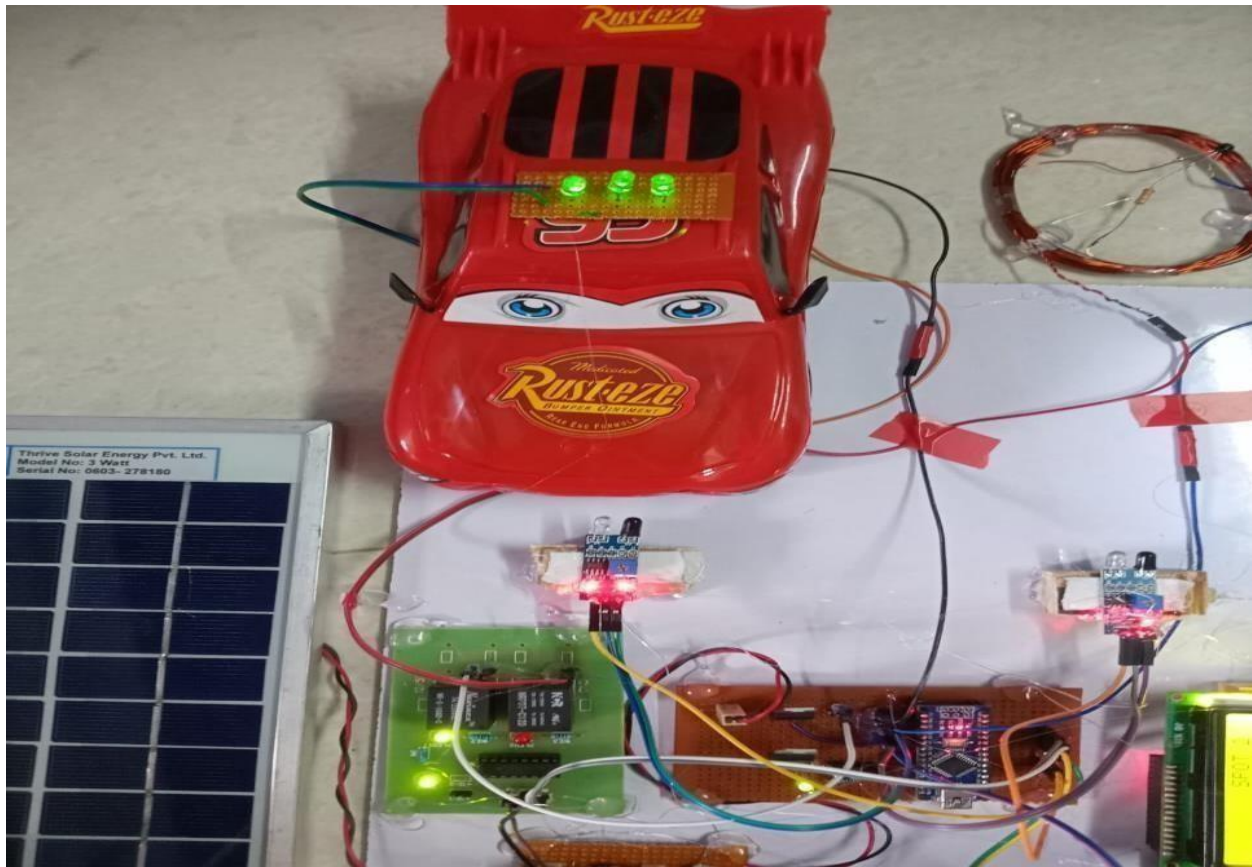


Fig 7.4: Slot 1 charging

- *This diagram provides a closer look at Slot 1 during the charging process.*
- *The primary coil in the charging station generates an electromagnetic field, which is received by the secondary coil in the EV to charge its battery.*
- *The LCD confirms that charging is active and will continue until the battery is full.*
- *The system ensures safe power transfer and prevents overcharging.*

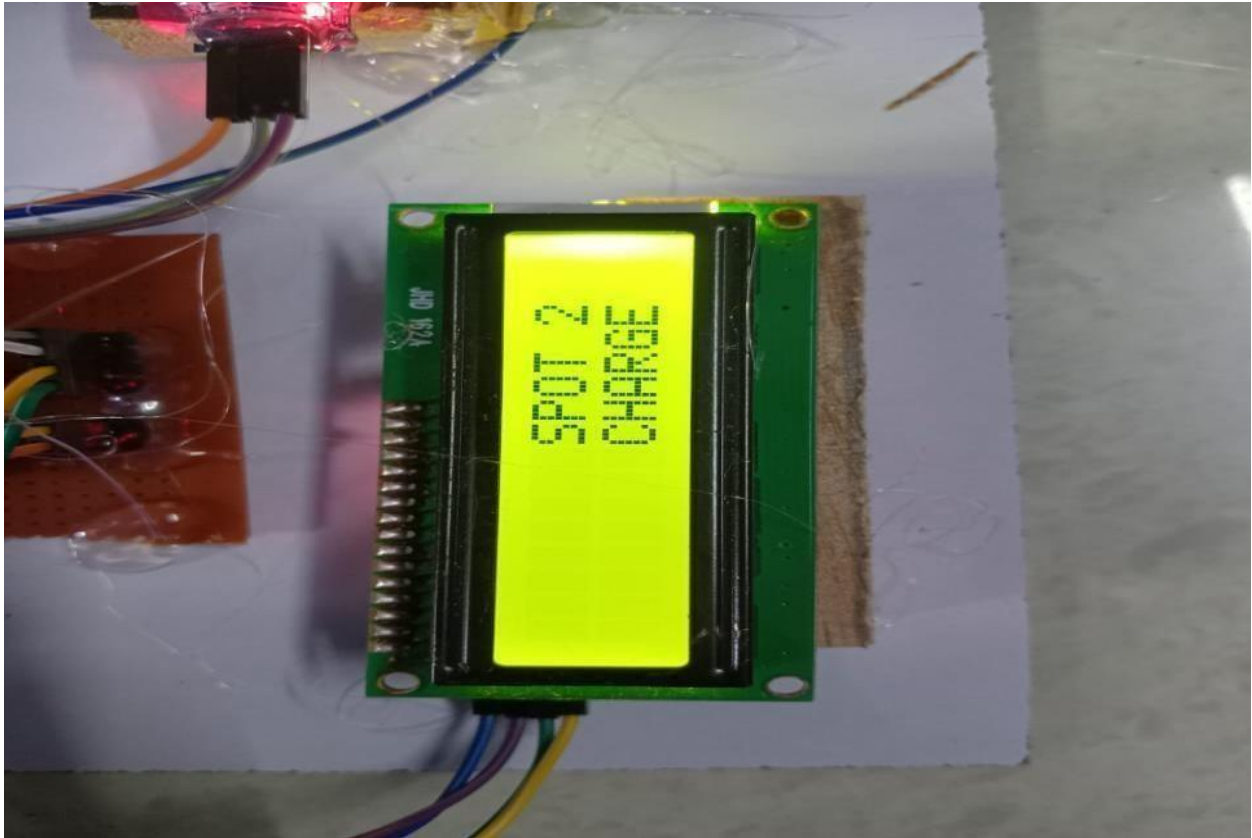


Fig 7.5 : Detected second slot for charging the car

- *This image represents the detection of a vehicle in Slot 2.*
- *Similar to Slot 1, the IR sensor detects the vehicle's presence and sends a signal to the Arduino.*
- *The LCD updates to show that Slot 2 is occupied, and the charging process begins.*
- *The system can independently operate both slots based on vehicle detection.*

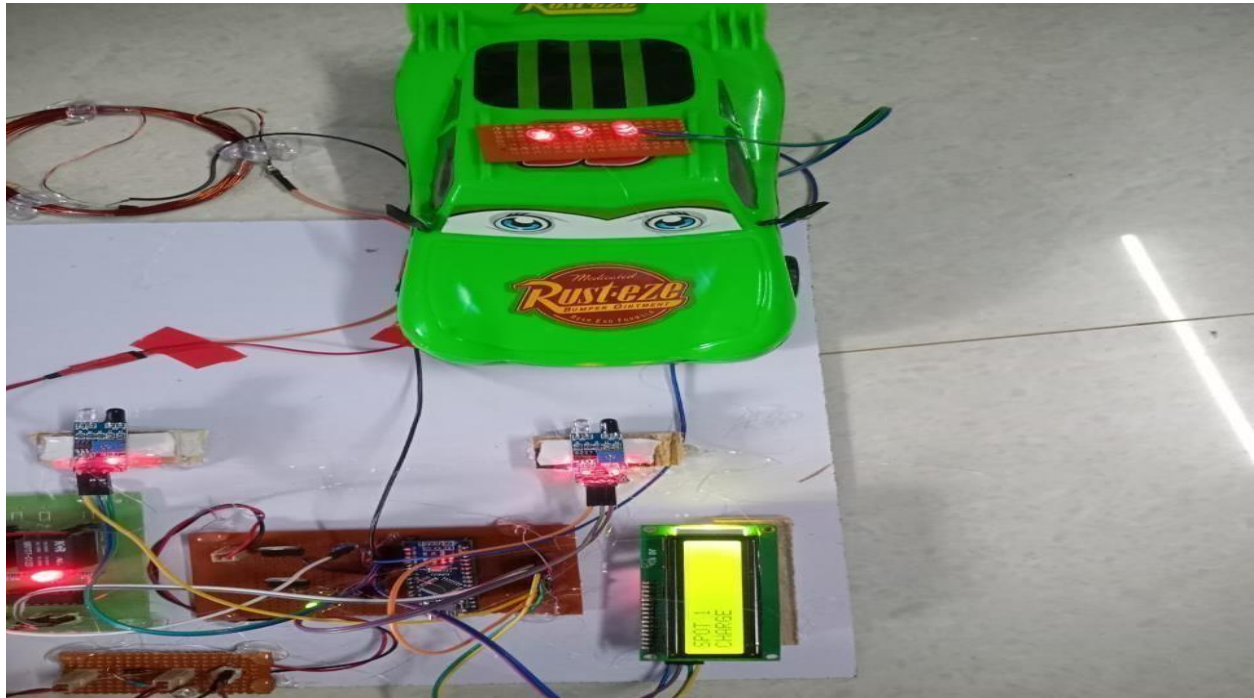


Fig 7.6 : Slot 2 charging

- *This diagram focuses on the charging process for Slot 2.*
- *The relay module is activated, allowing current to flow to the wireless charging coil for Slot 2.*
- *Wireless power transfer occurs through induction, enabling contactless charging.*
- *The LCD continues to update the status of the slot in real-time.*

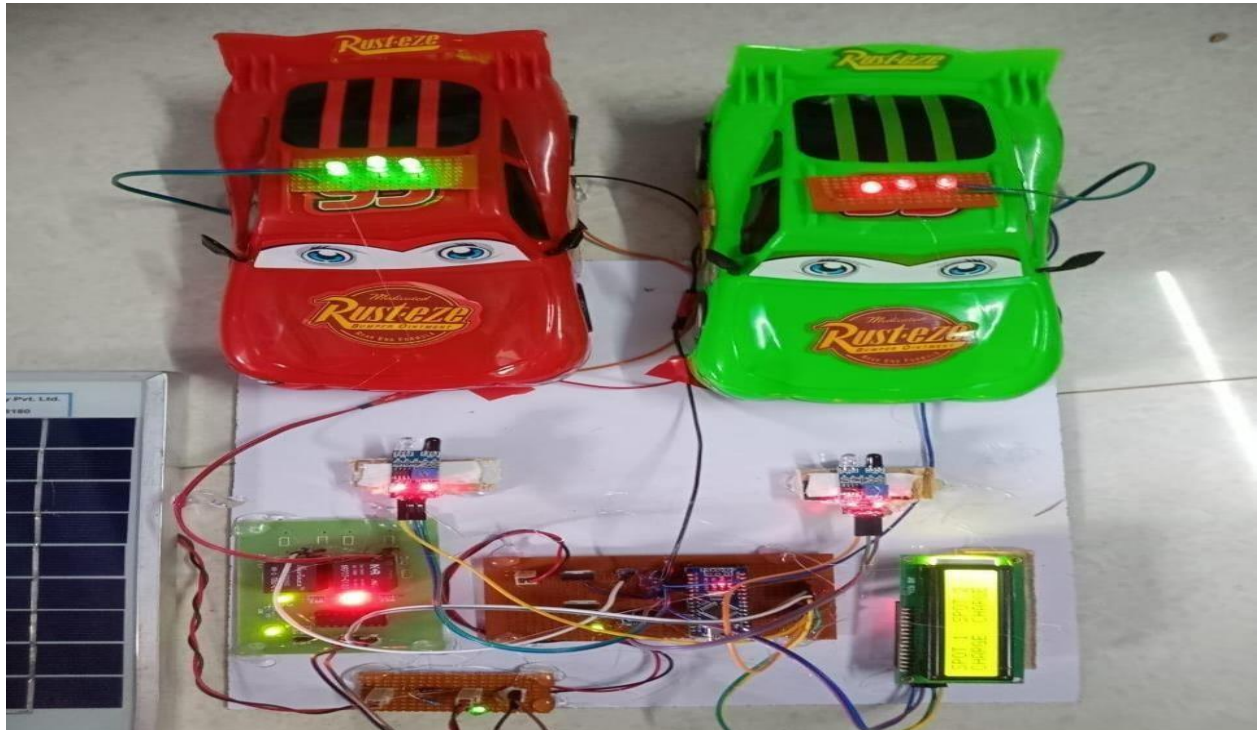


Fig 7.7 :2 Slots for at a time charging

- *This diagram represents a scenario where both slots are occupied and charging simultaneously.*
- *The system can handle multiple charging operations at once using separate relay modules and charging coils.*
- *Both slots operate independently, ensuring efficient power distribution and optimal charging.*
- *The LCD updates display messages like "Slot 1: Charging" and "Slot 2: Charging" to keep the user informed.*

CHAPTER-8

CONCLUSION

Conclusion

This project successfully demonstrates the development of a **wireless charging station for electric vehicles (EVs) using solar energy**. The implementation of **wireless charging technology** eliminates the need for conventional wired connections, making the charging process more efficient, convenient, and user-friendly.

Key Outcomes of the Project

1. Renewable Energy Utilization

- The system **harnesses solar energy** to power EVs, reducing reliance on **fossil fuels and conventional electricity grids**.
- This promotes **sustainable energy solutions** and helps in reducing **carbon emissions**.

2. Wireless Charging Technology

- By integrating **inductive power transfer**, the system enables **wire-free charging**, eliminating physical connectors and enhancing durability.
- This technology is already being implemented in **smartphones, electric vehicles (EVs), and industrial applications**, proving its effectiveness.

3. Automated & Smart Charging System

- The **IR sensors** detect vehicles in the charging slot, ensuring **automatic activation** of the charging process.
- **Arduino Nano controls** the system, making it **intelligent and efficient**.

4. Portability & Ease of Use

- The **absence of physical charging cables** ensures **minimal wear and tear**, reducing maintenance costs.
- Users can **simply park their vehicle**, and the system will begin **charging automatically**.

5. Real-World Applications & Scalability

- This project can be **expanded for commercial use**, especially in **public parking spaces, residential charging hubs, and smart city infrastructure**.
- The **scalability** of this solution makes it a promising candidate for **future urban transportation systems**.

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